A universal interatomic potential for solvation chemistry

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Solvation determines the thermodynamics, reactivity and transport of essentially every liquid-phase process. The need for conformational sampling in the liquid phase, the balance of ionic and dispersion interactions, as well as the possibility of bond breaking events, poses great challenges for the transferability of "universal" potentials, that aim to enable quantitative atomistic simulations across much of the periodic table. To assess quantitatively how existing potentials perform for this problem, we generate combinatorially MAD-SOL, a dataset of realistic solution configurations, covering 17 solvents, 78 cations, 35 anions, and compute energy and forces using a consistent level of electronic structure theory. We then compare a universal potential, PET-MAD-1.5, with one that is re-trained including also MAD-SOL, which we call PET-SOL, on a variety of structural and thermodynamic properties from the excess molar volume to pair correlation functions. The addition of solvation-specific data improves dramatically the test-set accuracy of the potential, allowing us to consider PET-SOL simulations as a proxy for the reference quantum method. We find that in most cases the general-purpose model yields results that are qualitatively correct, and usually in better agreement with the quantum targets than the quantum target is with available experiments. This paints a very positive picture for the field, in which universal models can be used with confidence to tackle subtle out-of-domain questions, and can be made much more accurate using a carefully curated but computationally affordable problem-specific dataset, which is especially useful when the underlying level of electronic structure theory is not sufficiently accurate for the problem at hand.

I. INTRODUCTION

From electrolytes and batteries to separations, homogeneous catalysis and pharmaceutical formulation, the behaviour of a solute in a liquid is set as much by its surrounding solvent as by the solute itself []. The quantities that matter in practice, coordination numbers, water-exchange rates, activity coefficients, ionic mobilities, are not properties of a single molecule but emergent properties of the first solvation shell and of its exchange with the bulk []. Molecular Dynamics (MD) simulations are an important tool for studying solvation on an atomistic level [1], however, resolving macroscopic observables demands statistics gathered over hundreds of atoms and nanoseconds, at an accuracy that still balances ion–solvent, solvent–solvent and dispersion interactions correctly. No established approach reaches that regime. On one side, classical force fields have the necessary reach, and optimized parameterizations now cover most of the periodic table, including the alkali, alkaline-earth, halide and lanthanide ions [2–9]. These models are, however, fitted almost exclusively to thermodynamic and structural targets, with water-exchange kinetics constrained for only a handful of ions [10, 11], and they treat polarization and charge transfer only effectively. A force field that reproduces the first-shell structure may therefore still misrepresent the exchange barriers that govern residence times. *Ab initio* molecular dynamics (AIMD) avoids these assumptions by treating ion–water polarization and many-body effects explicitly, but it is typically limited to a few hundred atoms and tens to hundreds of picoseconds, suffering from insufficient sampling or finite size effects [12]. Data-driven many-body potentials fitted to coupled-cluster references offer a remarkably accurate alternative: building on MB-pol for water [13], the MB-nrg potentials of Paesani and co-workers now describe the hydration structure, vibrational spectra and hydration free energies of the alkali-metal and halide ions [14–17], the distribution of halides at the air/water interface [18] and even solvated peptides [19], see Ref. 20 for a recent review. Because each new species, together with its two- and three-body interactions with every partner, has to be fitted individually, and molecular identities are fixed, this route has so far been restricted to a handful of chemically well-defined, non-reactive systems. Machine-learned interatomic potentials (MLIPs) promise both at once, and recent work follows two distinct routes. The first fits a dedicated, high-accuracy potential to an individual salt [21–23], usually from trajectory snapshots of that system: accurate where it was trained, but one system at a time. The second trains a single universal model over a broad chemical space — PET-MAD [24, 25], MACE-MP-0 [26], and more recently MACE-POLAR-1 [27], which adds explicit long-range electrostatics and was shown to describe several solvated systems. Universal machine-learned interatomic potentials (uMLIPs) are transferable by construction, but their training sets contain comparatively little dense, highly coordinated liquid, and solvation is not the regime they were optimized for. Whether that omission actually limits them is an open question, and it is the one this work sets out to answer: does describing solvation chemistry require solvation-specific training data, or is chemical diversity

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alone enough?

To answer it we build MAD-SOL, a dataset that treats solvation combinatorially: 78 cations and 35 anions combined with 17 solvents, giving 38 909 condensed-phase configurations that span 80 elements, computed at the r^{2}SCAN [28, 29] level with settings identical to those of the parent MAD-1.5 [25] dataset so that the two can be merged into a single training pool. Fine-tuning the PET-MAD-1.5 foundation model, built on the PET architecture [30], on that pool yields PET-SOL, released in a production (S) and a lightweight (XS) size and, for systems where dispersion dominates, with an optional Grimme-D3 correction [31].

Measured against the reference electronic structure, the added data improves the accuracy substantially. On solvated configurations PET-SOL is six times more accurate in energy and twice in forces than the foundation model, while remaining as accurate on the universal chemical space. Despite this, model differences in simulation matter much less. We compare models across a scan of solvation structure and dynamics of 32 monatomic ions spanning the alkali, alkaline-earth, halide and lanthanide series. The infinite dilution limit, in which a single ion only interacts with water, provides a natural setup for comparing ion-solvent interactions. We compare the base model and solvation specific PET-SOL on structural parameters, described by the Radial Distribution Function (RDF), and dynamical properties such as the mean residence time (MRT) of water molecules within the first solvation shell. This part of the work not only validates model accuracy, but also acts as a systematic, consistent analysis of ion solvation across chemical space. We further benchmark the models on macroscopic observables at finite concentration for 1 M NaCl, CsCl and $CaCl_2$ solutions. We compare solvation structure, diffusivities, activity derivatives and lyotropic behaviour via self-diffusion and water-reorientation dynamics. In order to assess performance on more complex ionic systems, we also compare densities and viscosities of the deep eutectic solvents (DESs) reline and ethaline. Further, we compute densities and viscosities of 17 common solvents spanning polar protic (water, methanol, ethanol, isopropanol, acetic acid, ammonia), polar aprotic (DMSO, DMF, ethylene carbonate, ethyl methyl carbonate, THF) and apolar or weakly polar (hexane, benzene, toluene, CCl_4, $CHCl_3$, diethyl ether) liquids, to test transferability beyond aqueous environments. Across these tests, both PET-MAD-1.5-S and PET-SOL-S reproduce first-shell structure and residence times across the ion series. To our knowledge, the 32-ion scan provides the first consistent *ab initio*-based map of hydration structure and water-exchange kinetics across the alkali, alkaline-earth, halide and lanthanide series, and the benchmarks beyond the dilute limit show that PET-SOL is considerably more certain and accurate on most macroscopic observables. We show that inclusion of dispersion effects is crucial for simulating solvents, and that PET-SOL-S + D3 improves results on deep eutectic solvents compared to the base PET-MAD-1.5-S + D3. Finally, we test the models on computing the autoionization constant of water, requiring a good description of O–H bond breaking and proton transfer. We can show that the three models underestimate pK_w, and that fine-tuning PET-SOL on a hybrid revPBE0 level allows to match experimental pK_w.

The paper is organized as follows. Section II A describes the design of the MAD-SOL chemical space, Sec. II B the reference calculations and Sec. II C the training of PET-SOL, while the simulation and analysis protocols are given in Sec. II D. The results open with the scan of single ions at infinite dilution (Sec. III A), followed by 1 M electrolyte solutions (Sec. III B), neat solvents (Sec. III C), deep eutectic solvents (Sec. III D) and the autoionization of water (Sec. III E). Section IV summarizes our conclusions.

II. METHODS

A. Design of the chemical space

In MAD-SOL, we treat solvation as a combinatorial problem: rather than sampling a small number of well-studied electrolytes in depth, we systematically combine a broad library of ions with a diverse library of solvents and sample the resulting systems. The library comprises 78 cations and 35 anions (Table I) and 17 solvents (Table II).

a. Solvated ions The cations span 69 monatomic species with formal oxidation states from +1 to +7: the alkali and alkaline-earth metals, the 3*d*–5*d* transition metals, the complete lanthanide and the actinides series, together with nine molecular cations, among them the hydronium and ammonium ions, two tetraalkylammonium ions and three imidazolium ionic-liquid cations. The anions comprise the four halides and 31 polyatomic species with charges from −1 to −3, spanning simple oxoanions, the carboxylates, the phosphate and sulfur series, the weakly coordinating fluorinated anions and the transition-metal oxoanions. The solvents cover protic and aprotic, polar and apolar, and battery-relevant classes.

For every cation–anion pair with formal charge equal to q_+ and q_- respectively, we select the smallest charge-neutral stoichiometry (m, n) so that $m\,q_+ + n\,q_- = 0$ and every configuration in the dataset is charge neutral. Each system is then built as a periodic cubic box subject to a hard budget of 150 atoms, a compromise between accommodating a genuine solvation shell and keeping DFT computational effort tractable across tens of thousands of configurations. Realised system sizes therefore range from 131 to 150 atoms. The cell edge is fixed from the experimental solvent density ρ at ambient conditions. Coordinates are generated with Packmol [32], which places the ions and the solvent molecules at random positions and orientations inside the box subject to a 2.0 Å minimum-distance tolerance, with periodic bound-

TABLE I. Cations and anions in the MAD-SOL library, grouped by the formal charge used to construct neutral formula units.

Formal charge	Species
+1	Ag, Cs, Fr, K, Li, Na, Np, Pu, Rb
+2	Ba, Be, Ca, Cd, Co, Cu, Hg, Mg, Mn, Ni, No, Pd, Pt, Sr, U, Zn
+3	Ac, Am, Au, Bk, Ce, Cf, Cm, Cr, Dy, Er, Es, Eu, Fe, Fm, Gd, Ho, Ir, La, Lu, Md, Nd, Pm, Pr, Rh, Ru, Sc, Sm, Tb, Tm, V, Y, Yb
+4	Hf, Os, Th, Ti, Zr
+5	Nb, Pa, Ta
+6	Mo, W
+7	Re, Tc
+1	H_3O^+, NH_4^+, $N(CH_3)_4^+$, $N(C_2H_5)_4^+$, $C_5H_5NH^+$, $CH_6N_3^+$, $C_6H_{11}N_2^+$ ([Emim]$^+$), $C_8H_{15}N_2^+$ ([Bmim]$^+$), $C_{10}H_{19}N_2^+$ ([Hmim]$^+$)
−1	F^-, Cl^-, Br^-, I^-, OH^-, CN^-, OCN^-, SCN^-, NO_3^-, $HCOO^-$, CH_3COO^-, $C_6H_5COO^-$, $CH_3SO_3^-$, $CF_3SO_3^-$, HCO_3^-, $H_2PO_4^-$, ClO_2^-, ClO_3^-, ClO_4^-, BrO_3^-, IO_3^-, MnO_4^-, BF_4^-, PF_6^-
−2	CO_3^{2-}, SO_4^{2-}, HPO_4^{2-}, $S_2O_3^{2-}$, $S_4O_6^{2-}$, S_2^{2-}, CrO_4^{2-}, $Cr_2O_7^{2-}$, MoO_4^{2-}, WO_4^{2-}
−3	PO_4^{3-}

TABLE II. Solvents in the MAD-SOL library. N_{at} is the number of atoms per molecule and ρ is the mass density from PubChem used to set the box volume of the NVT simulations (not used as an experimental target; see notes for the entries not at room temperature).

Solvent	IUPAC name	PubChem CID	Formula	N_{at}	ρ (g cm^{-3})
Water	oxidane	962	H_2O	3	0.9950
Ammonia	azane	222	NH_3	4	0.6819[a]
Methanol	methanol	887	CH_4O	6	0.7866
Ethanol	ethanol	702	C_2H_6O	9	0.7893
Propan-2-ol	propan-2-ol	3776	C_3H_8O	12	0.7851
Acetic acid	acetic acid	176	$C_2H_4O_2$	8	1.0446
N,N-Dimethylformamide	N,N-dimethylformamide	6228	C_3H_7NO	12	0.9500
Dimethyl sulfoxide	methylsulfinylmethane	679	C_2H_6OS	10	1.1000
Tetrahydrofuran	oxolane	8028	C_4H_8O	13	0.8833
Diethyl ether	ethoxyethane	3283	$C_4H_{10}O$	15	0.7134
Ethylene carbonate	1,3-dioxolan-2-one	7303	$C_3H_4O_3$	10	1.3214[b]
Ethyl methyl carbonate	ethyl methyl carbonate	522046	$C_4H_8O_3$	15	1.0130
Benzene	benzene	241	C_6H_6	12	0.8756
Toluene	toluene	1140	C_7H_8	15	0.8600
Hexane	hexane	8058	C_6H_{14}	20	0.6606
Chloroform	chloroform	6212	$CHCl_3$	5	1.4840
Tetrachloromethane	tetrachloromethane	5943	CCl_4	5	1.5940

[a] Density at the boiling point 240 K.
[b] Density at 313 K.

ary conditions enabled.

Enumerating $78 \times 35 \times 17$ gives 46 410 nominal solute/solvent systems, of which 46 321 (99.8 %) were successfully constructed; the remaining 89 are high-charge cations whose counter-ion stoichiometry leaves no room for solvent within the atom budget defined above. Each structure is then relaxed with the PET-MAD-1.5-S universal machine-learning interatomic potential (uMLIP) [25], evaluated through its dedicated `metatomic` ASE calculator [33, 34]. Each relaxed system is finally propagated with PET-MAD-1.5-S in the NPT ensemble using the isotropic Martyna–Tobias–Klein integrator [35] as implemented in ASE [34], with a thermostat damping time of 50 fs and a barostat damping time of 500 fs. Initial velocities are drawn from a Maxwell–Boltzmann distribution at the target temperature. The timestep is set to 0.5 fs and each trajectory is propagated for 10 ps, with configurations, energies, forces and the virial stress recorded every 50 fs. Rather than fixing a single thermodynamic state, each system is assigned its own randomly drawn temperature and pressure: T is sampled uniformly from 300–800 K and p uniformly from $\{0.1, 0.2, \ldots, 1.0\}$ GPa. This spreads the dataset over a broad range of densities and degrees of thermal disorder at no additional cost and includes compressed, superheated liquid states that a purely ambient dataset would never visit. From each trajectory, 10 configurations were randomly drawn, giving a pool of 463 210 PET-MAD-labelled configurations. This pool was then reduced to one configuration per unique cation/anion/solvent sys-

tem by farthest-point sampling (FPS) [36] in the PET-MAD-1.5-S feature space, retaining for each system the most diverse of its ten candidates. The resulting 46 321 configurations constitute the set submitted to DFT.

Together with these structures, two auxiliary subsets have been generated beyond neutral solvated-ion systems.

b. Solvents interface Solvent interfaces were generated for all 136 unique binary solvent combinations. For each mixture, boxes of the two component solvents at experimental density were stacked on top of one another in an cubic box, with the number of molecules of each component chosen to give similar volumes at experimental density such that the total system did not exceed 200 atoms. In addition, each of the 17 solvents was simulated individually at a vacuum interface. Following a 20 ps NVT equilibration at 350 K, a 50 ps NVT production run and a 50 ps NPT simulation at 350 K and 1 atm were run for each system with PET-MAD-1.5 in i-PI [37]. For the solvent-vacuum interface system, the NPT was replaced by a NVT. For all NVT simulations a generalized Langevin thermostat [38] was used. For NPT simulations we used the same thermostat with a Bussi-Parinello barostat [39]. Snapshots were taken every 1 ps. Five maximally diverse frames per trajectory were selected by FPS in a PET-MAD-1.5 feature space, yielding $(136+17)\times5 = 765$ structures.

c. Solvents soup In order to sample more diverse solvent-solvent interactions, replica-exchange MD was employed on a single periodic box containing one molecule of each of the 17 solvents (174 atoms), sampled both in the bulk and at a liquid-vacuum interface. The bulk system was simulated in a cubic box at the mean of the solvent's experimental densities (box length 12.87 Å). The interface system was built from the same composition by first assembling a cubic box at experimental density before adding vacuum along one dimension, giving an elongated $14 \times 14 \times 100$ Å cell. Both cells were held fixed throughout sampling in 100 ps of NVT production. Rather than sampling a single thermodynamic state, each system was sampled by temperature replica-exchange molecular dynamics (REMD) [40] with PET-MAD-1.5-S in i-PI [37], using 20 replicas geometrically spaced between 100–1000 K, exchanges between neighbouring replicas attempted every 50 fs, and a combined colored-noise/stochastic-velocity-rescaling thermostat [38, 41] on each replica; the timestep was 0.5 fs and each replica was propagated ≈100 ps. Configurations were subsampled every 2 ps from each replica and pooled across the ladder, giving ∼1000 candidates per cell (2000 in total, 1000 bulk and 1000 interface).

B. DFT calculations

All reference calculations were performed with FHI-aims (version 250806), an all-electron electronic-structure code based on numeric atom-centred orbitals [42]. The input parameters are standardised across the dataset and are, setting for setting, those used for MAD-1.5-[25]: the two datasets are computed at the same level of theory with the same numerical conventions, and are therefore fully consistent with one another, which is what allows them to be merged into the single training pool of Sec. II C without introducing any relative offset between their reference energies.

As the exchange–correlation functional we use the meta-GGA r^{2}SCAN [28, 29]. It improves accuracy significantly with respect to the GGAs used in most available DFT datasets, while keeping the computational cost affordable. Two of its documented advantages bear directly on solvation chemistry: it substantially improves formation enthalpies across a wide range of solids [43, 44] and, like SCAN, it describes far better than GGAs hydrogen bonding [45–47], the interaction that organizes the first solvation shell and, with it, most of the structural observables this dataset is meant to support. Robust convergence is enforced with an $8\,\text{Å}^{-1}$ k-point density grid for periodic systems and a Gaussian electronic smearing of 0.05 eV, together with self-consistency thresholds of 10^{-6} eV on the total energy, $10^{-4}\,\text{eV Å}^{-1}$ on the forces and $10^{-5}\,e\,a_0^{-3}$ on the electron density. Each calculation yields the total energy and the atomic forces.

The released converged dataset comprises 38 909 configurations totalling 5 746 889 atoms, each carrying r^{2}SCAN energies and forces. Its composition is summarised in Table III. Element coverage is 80 elements in the condensed-phase configurations and maximum force components remain below $20\,\text{eV Å}^{-1}$ throughout, indicating that the uMLIP-driven sampling did not generate pathological close contacts.

C. Model training

Two potentials are released alongside the dataset, differing only in capacity: a production model, PET-SOL-S, and a lightweight model, PET-SOL-XS. Both are obtained by fine-tuning the corresponding PET-MAD-1.5 foundation models [25] on the union of the parent MAD-1.5 dataset and MAD-SOL. Both models use the point-edge transformer (PET) architecture [30] exactly as distributed with PET-MAD-1.5; the fine-tuning does not add, remove or resize any layer, so the parameter count is unchanged from the foundation model and every weight is inherited. A single architectural hyperparameter was changed before fine-tuning: PET-MAD-1.5 uses an *adaptive-cutoff* scheme: instead of taking every neighbour inside the fixed radial cutoff r_c, each atom i is assigned its own cutoff $r_i \le r_c$ chosen so that it has approximately n_{adapt} neighbours. The released PET-MAD-1.5 models use $n_{\text{adapt}} = 8$ (XS) and 16 (S), values appropriate to the largely crystalline and molecular environments. Dense solvated systems are more highly coordinated: the first solvation shell of a multiply charged cation, together

TABLE III. Composition of the released MAD-SOL dataset. $\max|F|$ is the largest force component encountered in the subset. The `solvents_soup` entry is a single composition sampled in bulk and at a vacuum interface.

Subset	Configurations	Atoms/config.	Atoms	Distinct systems	Elements	$\max\|F\|$ (eV Å^{-1})
`solvated_ions`	36 144	131–150	5 253 234	36 144	80	19.9
`solvents_soup`	2 000	174	348 000	1	6	11.8
`solvents_interface`	765	171–200	145 655	153	6	7.2
Total	38 909	148	5 746 889		80	

with its counter-ions and the surrounding solvent, contains substantially more atoms than that, so the inherited values truncate the atomic environment well inside the nominal cutoff sphere. We therefore raised n_{adapt} to 40 for both models before any training, which is a change to the neighbour-selection rule alone and leaves the weights untouched. The relevant hyperparameters are collected in Table IV.

TABLE IV. Architecture of the two released models. Both inherit every hyperparameter from the corresponding PET-MAD-1.5 foundation model except the adaptive neighbour count n_{adapt}, which was raised to 40 before fine-tuning (values in parentheses are those of the original foundation model).

Hyperparameter	XS	S
Radial cutoff r_{c} (Å)	7.5	8.0
Adaptive neighbour count n_{adapt}	40 (8)	40 (16)
Edge feature dimension	128	256
GNN layers	2	3
Attention layers	1	1
Parameters	4.5M	25.9M

Fine-tuning solely on MAD-SOL would specialize the models onto solvated systems at the cost of the general-purpose coverage that makes PET-MAD-1.5 useful as a starting point. Both models are therefore fine-tuned on the union of MAD-1.5 and MAD-SOL, so that solvation chemistry is added to, rather than substituted for, the parent chemical space. The combined pool holds 255 712 configurations, of which MAD-SOL contributes 38 909 (15.2 %); because the solvated systems are an order of magnitude larger than the average MAD-1.5 structure, they nevertheless account for 63.7 % of its 9 018 933 atoms. The MAD-SOL pool is divided 80:10:10 into training, validation and test sets within each constituent subset and added to the corresponding released splits of MAD-1.5 (Table V).

The learning target is the atomization energy, computed subtracting the isolated atoms energies from the total energy, together with its positional gradient. Both models are fully fine-tuned — every weight is trainable, no layer is frozen and no adapter is introduced — for 250 epochs on energies and forces, minimizing a weighted sum of mean-squared errors evaluated on standardized targets, with the per-element baseline and the scaling factors recomputed from the training split; the remaining settings are collected in Table VI. The fine-tuning was repeated with four random seeds for each model size, which changed the validation MAEs by less than 1 %.

TABLE V. Composition of the training, validation and test splits. Configuration counts are given per split; the last column is the total number of atoms contributed by each dataset. The 102 isolated-atom configurations are counted with MAD-1.5 and, like its dimer and trimer subsets, are assigned entirely to the training split.

Source	Train	Validation	Test	Total	Atoms
MAD-SOL	31 127	3 890	3 892	38 909	5 746 889
MAD-1.5	180 184	18 305	18 314	216 803	3 272 044
Total	211 311	22 195	22 206	255 712	9 018 933

TABLE VI. Optimisation settings. Weights are those entering the total loss; the forces are obtained by differentiating the predicted energy with respect to the atomic positions.

Setting	Value
Initial weights	PET-MAD-1.5, $n_{\mathrm{adapt}} = 40$
Fine-tuning	full (all weights trainable)
Epochs	250
Targets	energy, forces
Loss weights	energy 10, forces 1
Learning rate	10^{-4}, cosine annealed
Batch size	256 (S), 512 (XS)

Table VII reports the errors of the two released models on the MAD-SOL and MAD-1.5 test sets, against two references: the unmodified PET-MAD-1.5 foundation models, and PET-MAD-1.5-S-40nn, a control obtained by applying the identical recipe — same architecture with $n_{\mathrm{adapt}} = 40$, same optimizer, same schedule, same number of epochs — to MAD-1.5 alone. The control separates the contribution of the MAD-SOL data itself from that of the larger neighbor count and of the additional optimization.

On MAD-SOL the fine-tuning reduces the energy MAE of the foundation model by a factor of six and the force MAE by a factor of two for the S size, and by factors of four and two for XS. Much of the foundation models' error there is a systematic offset rather than scatter: they underbind by 8.0 (S) and 15.0 (XS) meV atom^{-1} on average, against 0.1 and 0.3 meV atom^{-1} for the released models. Removing that offset still leaves them at 6.0 and 8.2 meV atom^{-1}, so the fine-tuning corrects both

TABLE VII. Accuracy of the released models on the two held-out test sets, compared with the PET-MAD-1.5 foundation models and with the MAD-only control PET-MAD-1.5-S-40nn, i.e. the S foundation model with $n_{\mathrm{adapt}} = 40$ fine-tuned by the identical recipe on MAD-1.5 alone. Energies are per atom, referred to the atomization energy; forces are per Cartesian component.

	MAD-SOL test (3892 config.)		MAD-1.5 test (18 314 config.)	
	E (meV atom^{-1})	F (meV Å^{-1})	E (meV atom^{-1})	F (meV Å^{-1})
Model	MAE (RMSE)	MAE (RMSE)	MAE (RMSE)	MAE (RMSE)
PET-SOL-S	1.37 (2.19)	19.6 (46.2)	10.48 (44.22)	38.2 (79.3)
PET-MAD-1.5-S-40nn	5.69 (9.31)	39.1 (94.2)	10.48 (45.10)	36.3 (75.7)
PET-MAD-1.5-S	8.14 (11.41)	42.9 (97.9)	11.09 (45.69)	36.8 (68.6)
PET-SOL-XS	3.45 (5.20)	38.1 (77.5)	15.31 (52.50)	67.0 (125.3)
PET-MAD-1.5-XS	15.10 (18.54)	74.9 (138.2)	18.73 (54.38)	86.4 (148.1)

the energy scale and a genuine loss of accuracy. The control recovers roughly half of the gap — 5.7 against 8.1 meV atom^{-1} for the S size — which is the part attributable to the larger neighbor count and the additional optimization; the remainder, down to 1.4 meV atom^{-1}, is what the MAD-SOL data itself contributes. On the parent MAD-1.5 test set PET-SOL-S matches both the control and the foundation model to within 0.6 meV atom^{-1}, so nothing is lost by adding solvation chemistry to the training pool. The energy RMSE exceeds the MAE by a large factor on the full set while the two are comparable on MAD-SOL, which reflects a small number of high-error outliers in the more heterogeneous parent dataset rather than a systematic failure. Parity plots for all five models on both test sets are collected in the Supplemental Material V. Given that the increase to 40nn is crucial for model performance on condensed phase systems, we use the PET-MAD-1.5-S-40nn variant as comparison to PET-SOL-S, and for brevity simply refer to the former as PET-MAD-1.5-S.

D. Molecular Dynamics Simulations

1. General simulation protocol

If not stated explicitly, all production simulations were run with LAMMPS [48] with the released models loaded through the `metatomic` pair style [33]. The equations of motion were integrated with a timestep of 0.5 fs. NVT simulations use a canonical-sampling-velocity-rescaling thermostat [41] at 330 K and a damping time of 50 fs. NPT simulations use a Nosé–Hoover thermostat and isotropic barostat [35] at 330 K and 1 bar with damping times of 50 fs and 500 fs, respectively. Simulations were run at increased temperatures to reproduce the structure of water at ambient conditions [46, 49]. Experimental references are taken at room temperature unless stated otherwise.

2. Single ion simulations

The single-ion scan and FeCl series share the same six-stage protocol. Initial configurations were generated with Packmol [32] and subsequently relaxed by conjugate-gradient energy minimisation. 10 ps NVT equilibration is followed by a 150 ps NPT run to sample the equilibrium density. The first 50 ps are discarded and the remaining 100 ps are averaged to obtain the mean cell volume. The box is isotropically rescaled to that mean volume over a further 2.5 ps of dynamics. A second, 10 ps NVT equilibration relaxes the system at the fixed, rescaled volume. Production is run for 5 ns in NVT and used for all of the structural and dynamical analysis below, with configurations saved every 50 fs.

For the ion scan, each of the 32 ions was placed in a cubic box with 256 water molecules, without counter ion with initial volume chosen to match the model's density of water. For the Fe–Cl series, a single Fe was placed with the respective number of chloride ions.

3. Finite-concentration electrolyte simulations

Beyond the single-ion boxes of Sec. II D 1, $CaCl_2$, NaCl and CsCl were additionally simulated at finite concentration ($\approx$ 1 M) with PET-SOL-S + D3, PET-SOL and PET-MAD-1.5-S, using the same LAMMPS/`metatomic` setup. Initial configurations were built with (Packmol [32]) placing ions in 25 Å cubic boxes at the expected molarity after NPT equilibration, correcting for the offset caused by the model's own equilibrium density (Sec. II D 3). The 25 Å boxes contain the corresponding numbers of cations and anions at the salt stoichiometry (Table XII in the SI). In order to avoid metastable ion pairs, e.g. Ca-Cl pairs, the cation and anion are placed first, alone in the box, at a loose 5 Å tolerance. Water is then packed into the remaining free volume around the pre-placed ions at the standard 2 Å tolerance, allowed enough space to form solvation shells.

Four replicas were run for each ion and each model. Equilibration was done for 10 ps NVT, followed by 150 ps NPT equilibration at 330 K/1 atm. Equilibrium volumes

from the last 50 ps from each replica were averaged to obtain the mean volume $\langle V \rangle$. Isotropic rescaling of the box to $L_{\text{final}} = \langle V \rangle^{1/3}$ over 2.5 ps, 10 ps NVT re-equilibration at the fixed, rescaled volume, and NVT production for 2.5 ns for each replica. Production configurations were saved every 100 steps as for the dilute limit simulations. Velocities were collected every 2 fs.

4. *Bulk solvents*

For both bulk solvent simulations and solvent-in-solvent simulations in the dilute limit, a standard protocol was used throughout. Boxes of ≈ 1000 atoms were built at experimental density, minimised by conjugate gradients and equilibrated for 50 ps in the NVT ensemble. Densities were obtained from NPT production, ran until density was equilibrated ((**CMnote:** HOW LONG? GIVE A RANGE)). Viscosities were obtained from NVT runs at the equilibrium density ((**CMnote:** HOW LONG? GIVE A RANGE)).

5. *Deep eutectic solvents*

Density of reline and ethaline systems were obtained via NPT simulation with 3.5 ns of production time for each system. Densities were calculated by averaging the density for the last 2.5 ns of simulation. Standard errors were estimated by dividing this interval into 10 equally sized blocks and performing block averaging. The viscosity of reline and ethaline systems was then obtained by deforming the ending configuration from NPT simulations to the simulated density, performing 50 ps of NVT equilibration at this targeted density, and then 5.5 ns of NVT production with the off-diagonal components of the stress tensor sampled every 2 fs. Cepstral analysis was then performed using SporTran for final calculation of viscosities and the corresponding uncertainties [50].

6. *Water autoionization*

PET-MAD-1.5-S and PET-SOL-S, along with two fine-tuned versions of PET-SOL-S are used in the simulations of this section. The dataset of Ref. [51] is used for fine-tuning. This dataset was originally labeled with revPBE0-D3. In this paper, we relabeled it with r^{2}SCAN by FHI-aims with exactly the same setting up mentioned in section II B. Both datasets, labeled with revPBE0-D3 and labeled r^{2}SCAN, are used in the fine-tuning. The fine-tuning is conducted with `metatrain`, and the full fine-tuning scheme was used. The model fine-tuned with the r^{2}SCAN dataset reached a per-atom energy MAE (RMSE) of 0.45 (0.29) meV atom^{-1} and a force MAE (RMSE) of 27.31 (15.02) meV Å^{-1}. The model fine-tuned with the revPBE0-D3 dataset reached a per-atom energy MAE (RMSE) of 0.94 (0.64) meV atom^{-1} and a force MAE (RMSE) of 33.35 (20.50) meV Å^{-1}.

The autoionization and path integral simulations were run with i-PI [37] and the `metatomic` interface. The equations of simulations were integrated with a timestep of 0.5 fs. All simulations were run with the canonical-sampling-velocity-rescaling thermostat and the Bussi-Zykova-Parrinello barostat at 1 bar with damping times of 20 fs and 100 fs, respectively. All simulations were run at 330 K, except from the simulations that were run with the model fine-tuned on the revPBE0-D3 dataset, which were run at 300 K. The simulation systems contain 128 water molecules.

The simulation procedures are the same as described in Ref. [52]. First, metadynamics simulations are conducted, and bias potentials of a height of $1 k_{\text{B}}T$ and a width of 0.3Å are deposited every 1000 steps along the charge-separation coordinate s, which represents the distance between the hydronium and hydroxide as explained in Ref. [52]. The bias depositing was stopped after several dissociation-association events of the hydronium and hydroxide were observed, which means that a reasonable bias potential has been established, resulting in trajectories of at least 1.75 ns long. Ten independent runs were started with the frozen bias potential from structures sampled from the trajectory of the metadynamics run to get the free energy surface. Replica exchange molecular dynamics simulation is used to help the sampling. The target temperatures are exchanged between replicas. For the model fine-tuned on the revPBE0-D3 dataset, there are 12 replicas, and the target temperatures are 300 K, 314 K, 330 K, 344 K, 361 K, 378 K, 396 K, 415 K, 434 K, 456 K, 477 K, and 500 K, respectively. For all other models, there are 10 replicas, and the target temperatures are the same as the ones used in the model fine-tuned on the revPBE0-D3 dataset, while start from 330 K. At least 100 ps long simulations were done for each replica. Thermodynamic integral method was employed to promote the classical water autoionization constant to the quantum-corrected constant, where path-integral molecular dynamic simulations (PIMD) were conducted with the local Langevin thermostat attached to the centroid mode, where each atom was represented by a 32-bead ring polymer. The lengths of PIMD trajectories are at least 100 ps

E. Analysis

1. *Radial distribution functions*

Ion–O, ion–ion and O–O radial distribution functions (RDFs) were computed from the production trajectories with `MDAnalysis` [53].The potential of mean force (PMF) $W(r) = -k_{\text{B}}T \ln g(r)$ is the reversible work required to bring a water oxygen from the bulk to a distance r from the ion, averaged over all other degrees of freedom [54]. Its minimum lies at r_{min} and the barrier separating the

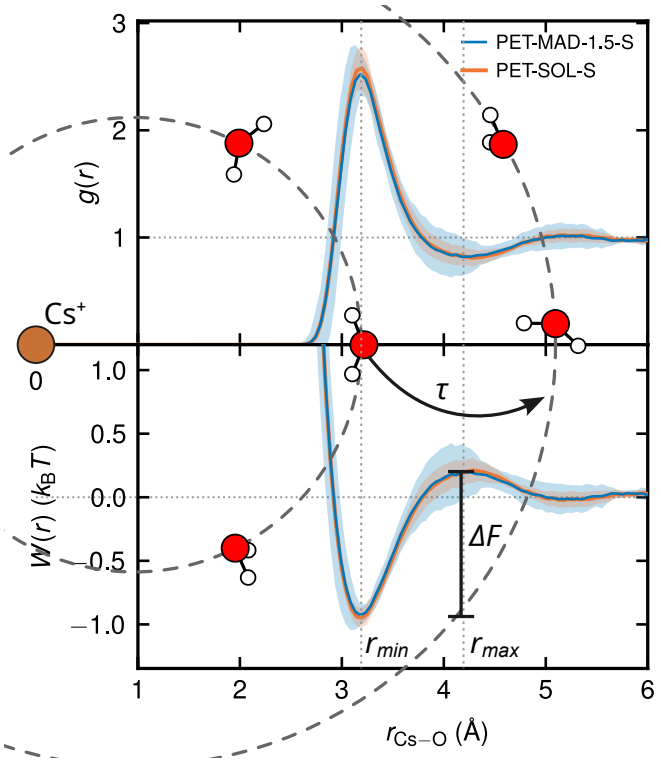


FIG. 1. Cs^+–O radial distribution function $g(r)$ (top) and potential of mean force $W(r) = -k_B T \ln g(r)$ (bottom) for PET-MAD-1.5-S and PET-SOL-S at 330 K. Shading indicates the LLPR committee uncertainty. (r_{max}, r_{min}, ΔF, r_c) are read off these curves for every ion. The arrow is showing a schematic path of a water molecule leaving the first solvation shell.

first from the second hydration shell at r_{max}, so that

$$\Delta F = -k_B T \ln \frac{g(r_{max})}{g(r_{min})} \tag{1}$$

is the free-energy barrier for leaving the first shell, as shown in Fig. 1. The first-shell coordination number is $CN = 4\pi\rho_O \int_0^{r_{min}} g(r)\, r^2\, dr$, with ρ_O the bulk number density of water oxygens.

2. *Uncertainty quantification*

Model uncertainties were estimated with the last-layer prediction rigidity (LLPR) committee [55] of 64 members. Trajectories were reweighted to each committee member [56], using a first order cumulant expansion of the PMF or densities. The uncertainties of ΔF, CN, RDF extrema or densities are the standard deviation over the 64 members of the LLPR committee, obtained by reweighting the RDF of the baseline model (SI-sec. V I) []; all quantities are evaluated at the r_{max} and r_{min} of the baseline RDF.

3. *Mean residence times*

Water residence times in the first hydration shell were computed from the survival function of Impey, Madden and McDonald [57]. For every water oxygen j, an indicator $h_j(t) \in \{0, 1\}$ records whether it lies within the shell cutoff r_c of the ion in frame t. The number of shell waters that survive a lag t,

$$n(t) = \left\langle \sum_j P_j(t_0, t; t^*) \right\rangle_{t_0}, \tag{2}$$

is averaged over all time origins t_0. Here $P_j(t_0, t; t^*) = 1$ if water j is inside the shell at t_0 and at $t_0 + t$ and has not left it for a continuous period longer than t^* in between, and $P_j = 0$ otherwise. Brief excursions across the cutoff therefore do not end a residence, while a departure that lasts longer than t^* counts as an exchange. We use $t^* = 2$ ps, as in the original definition [57] and in recent work on ion hydration [15]. By construction $n(0)$ is the mean number of waters inside r_c, and the normalised survival function $C(t) = n(t)/n(0)$ decays from one. The mean residence time is its integral,

$$\langle \tau \rangle = \int_0^{t_{max}} C(t)\, dt, \tag{3}$$

evaluated with the trapezoidal rule on 1000 logarithmically spaced lag times between 0 and the analysed trajectory length t_{max}. The logarithmic grid resolves the first picoseconds of the decay, which dominate the integral for the fast-exchanging ions. The shell cutoff r_c is determined for every ion and model from its own ion–O RDF (Sec. II E 1). We use the shell cutoff r_c at the inflection point of the running coordination number $n(r) = 4\pi\rho_O \int_0^r g(r')\, r'^2\, dr'$, i.e. the first minimum of $r^2 g(r)$, rather than simply the RDF's first minimum, following Ref. 14. This gives the correct radial free energy along the ion–O distance, dictating the waters exchange rate. For weakly hydrated ions such as Cs^+ or I^- the first minimum is shallow and broad, so that a cutoff placed there lies in a region where water molecules continually enter and leave the shell, whereas the inflection point is well defined for all ions. Residence times were computed from the same 5 ns NVT production trajectories. (Sec. II D 1), analysed at a frame spacing of 0.5 ps. Their uncertainty is the standard error over five consecutive 1 ns blocks. Like the absolute value of $\langle \tau \rangle$ itself, which depends on t^* and on the choice of r_c [58], these errors are only indicative for the slowest ions. For PET-SOL Be^{2+}, $C(t)$ does not decay appreciably within the trajectory, and only a lower bound of $\langle \tau \rangle > 2.5$ ns (half the trajectory length) is given (Table XI).

4. *Diffusion coefficients*

The coefficients $D(L)$ were obtained from the production runs via a straight line fit of the slope of the

mean-squared displacement of the respective atoms, taking only the linear part into account. Given the limited box size and artificial boundaries, a finite-size correction was applied [59]:

$$D(\infty) = D(L) + \frac{k_B T \zeta}{6\pi\eta L}, \tag{4}$$

where k_B is the Boltzmann constant, T the temperature, $\zeta \approx 2.837297$ a geometric factor for cubic boxes, η the shear viscosity, and L the box length. For η we use the pure-water viscosity simulated with the same model (Table IX). The uncertainty from the viscosity estimation is propagated to the diffusion coefficeints.

5. *Water reorientation*

Water reorientation was characterised by the second-order Legendre correlation function of the O–H bond unit vector $\mathbf{u}$, $C_2(t) = \langle P_2[\mathbf{u}(t_0)\cdot\mathbf{u}(t_0+t)]\rangle$ with $P_2(x) = (3x^2-1)/2$, treating the two O–H bonds of each water as independent [60]. Bond vectors were sampled every 0.1 ps, with time origins every 1 ps and lags up to 15 ps. At each time origin t_0, a water is assigned to the cation (anion) shell if its oxygen lies within the first minimum of the cation–O (anion–O) radial distribution function of the same system, and to bulk otherwise; the assignment is made only at t_0. For each population, $C_2(t)$ is accumulated over all O–H bonds and time origins of a replica before normalising. Each replica's $C_2(t)$ was fit with a bi-exponential, $Ae^{-t/\tau_1} + Be^{-t/\tau_2}$, and the reorientation time $\tau_{\mathrm{reor}} = A\tau_1 + B\tau_2$ is averaged over replicas. Ion effects are reported as $\tau_{\mathrm{reor}}^{\mathrm{shell}}/\tau_{\mathrm{reor}}^{\mathrm{bulk}}$, with the replica standard errors propagated. One PET-MAD-1.5-S NaCl replica, which contains an autoionised H_3O^+/OH^- pair from the start of production, was excluded from the reorientation analysis.

6. *Activity-coefficient derivative*

To study ion-ion association and the effect of solvation on the salt's thermodynamic activity, we compute the salt's thermodynamic (activity) factor $\Gamma \equiv \partial \ln a_\pm / \partial \ln c = 1 + \partial \ln \gamma_\pm / \partial \ln c$, which is obtained through the Kirkwood–Buff (KB) formalism [61]. For each species pair ij the KB integral

$$G_{ij} = \int_0^\infty 4\pi r^2 \left[g_{ij}(r) - 1\right] \mathrm{d}r \tag{5}$$

is computed from the ion–ion and ion–water RDFs (cation–cation, anion–anion, cation–anion, cation–O, anion–O) using `kbkit`. Because $g_{ij}(r)$ is only known up to half the (finite) box edge, Eq. 5 is evaluated with the analytical hypersphere finite-volume weighting of Krüger *et al.* [62, 63], applied to an RDF first corrected for the finite-system particle-number depletion of Ganguly and van der Vegt [64], following the general finite-volume KB framework reviewed in Ref. [65]. We report u_2 corrected KB integrals [63].

The KB integrals enter the activity derivative through the mean-ion-convention expression used for 1:1 electrolytes (NaCl, CsCl) by Weerasinghe and Smith [66] and Zhang *et al.* [67]:

$$\Gamma = \frac{\partial \ln a_\pm}{\partial \ln c} = \frac{1}{1 + \rho_{\mathrm{salt}}\,(G_{cc} - G_{co})}, \tag{6}$$

$$G_{cc} = \tfrac{1}{4}(G_{++} + G_{--} + 2G_{+-})\,, \tag{7}$$

$$G_{co} = \tfrac{1}{2}(G_{+O} + G_{-O})\,, \tag{8}$$

with ρ_{salt} the formula-unit number density. For the 2:1 electrolyte $CaCl_2$, G_{cc} and G_{co} are instead weighted by the ion mole fractions $x_c = \nu_c/\nu$, $x_a = \nu_a/\nu$ ($\nu_c = 1$, $\nu_a = 2$, $\nu = 3$), as used for $MgCl_2$ by Grotz *et al.* [68], following Refs. 66, 69:

$$G_{cc} = \tfrac{1}{9}[G_{++} + 4\,(G_{--} + G_{+-})]\,, \tag{9}$$

$$G_{co} = \tfrac{1}{3}G_{+O} + \tfrac{2}{3}G_{-O}, \tag{10}$$

with Γ still given by Eq. 8. Γ is computed on the molarity scale and converted to the molality scale of the experimental (Pitzer) reference as $\Gamma_m = \Gamma_c\,(1-w)\,\rho/\rho_{\mathrm{w}}$, where w is the mass fraction of salt, ρ the simulated solution density and ρ_{w} the density of the same model's pure water, which assumes a density linear in c between pure water and the simulated solution.

7. *Calculation of Bulk Solvent Density and shear viscosity*

Density equilibration was assessed from the time evolution of density across the trajectory using the PyMBAR [70]. Additional details on implementation are included in the SI along with retained trajectory lengths for each system. (**CMnote:** NATALIE PLESE ADD DETATILS HERE `DENSITY_EQUILIBRATION_ENFORCED_MIN_RETAIN.CSV`). The reported bulk density of each solvent was calculated as the mean of all instantaneous density observations in the retained, equilibrated trajectory. This retained portion of the trajectory was divided into 10 blocks of approximately equal size to calculate block means, $\bar{\rho}_i$, and estimate the standard error of the mean.

Shear viscosities were computed from the Green–Kubo relation [71, 72],

$$\eta = \frac{V}{k_{\mathrm{B}}T}\int_0^\infty \langle \sigma_s(t)\,\sigma_s(0)\rangle\,\mathrm{d}t = \frac{V}{2k_{\mathrm{B}}T}\,S(0), \tag{11}$$

where σ_s is any of the off-diagonal stress components σ_{xy}, σ_{xz}, σ_{yz} and $S(\omega)$ is their power spectrum. Rather than integrating the noisy autocorrelation function, we estimate $S(0)$ by cepstral analysis [73, 74] as implemented in `SporTran` [50]: the logarithm of the sample spectrum,

averaged over the three components, is low-pass filtered by retaining its first P^* Fourier (cepstral) coefficients, with P^* chosen by the Akaike information criterion. The resulting estimate of $\ln \eta$ is normally distributed, and its standard deviation, which is known analytically, gives the reported uncertainty.

F. Autoionization simulations

1. $\mathrm{p}K_\mathrm{w}$ *calculation*

The autoionization constant is obtained through the potential of mean force $W(r)$:

$$K = \frac{1}{c^{\circ 2} V} \left\{ \int_0^{r_\mathrm{c}} 4\pi r^2 \exp\left[-\beta \left(W(r) - C\right)\right] \mathrm{d}r \right\}^{-1}, \tag{12}$$

where $r_\mathrm{c} = 7\mathring{A}$, $\beta = 1/k_\mathrm{B}T$, c° is converted into $1660.5391\,\mathrm{\AA}^{-3}$, V is the size of the simulation box, and C is a constant for removing the Coulomb interaction between the hydronium and hydroxide, calculated in the same way as defined in Ref. [52].

$W(r)$ is accessible from the free energy curve $G(s)$, which is calculated through reweighting:

$$G(s) = -\frac{1}{\beta} \ln \frac{\left\langle \delta\left(s - s(\mathbf{R})\right) e^{\beta V_\mathrm{bias}(s(\mathbf{R}))} \right\rangle_\mathrm{bias}}{\left\langle e^{\beta V_\mathrm{bias}(s(\mathbf{R}))} \right\rangle_\mathrm{bias}}, \tag{13}$$

where $\langle\cdot\rangle_\mathrm{bias}$ indicates an average over the biased ensemble, $\delta(\cdot)$ is the Dirac delta, s denotes the charge-seperation coordinate, and $\mathbf{R}$ represents the positions of the atoms in the simulation system. The free energy curve is converted into the potential of mean force by

$$W(r) = G(r) + \ln\left(4\pi r^2\right), \tag{14}$$

where we replace the charge-separation coordinate s with the radial distance r between the hydronium and hydroxide.

2. *Nuclear quantum effects correction*

Thermodynamic integration allows us to calculate the quantum correction to the classical free energy inexpensively through:

$$\Delta\Delta G_\mathrm{NQE} = 2 \int_0^1 \frac{\langle T \rangle^\mathrm{D}_{1/y^2} - \langle T \rangle^\mathrm{A}_{1/y^2}}{y} \,\mathrm{d}y, \tag{15}$$

where D and A stand for the dissociated and associated states, respectively, and $y^2 = m^\mathrm{phys}/m$, representing the ratio of the real particle mass and the simulation particle mass.

III. RESULTS

A. Systematic exploration of monoatomic ions in water

Each of the 32 monatomic ions of Table I, spanning the alkali, alkaline-earth, halide and lanthanide series, was simulated as a single ion in a periodic box of water without a compensating counter-ion. For every ion we obtain the ion–O RDF and PMF and from them the first-shell barrier ΔF (Eq. (1), Fig. 1), together with the mean residence time $\langle\tau\rangle$ of water in the first shell (Sec. II E 3). All values are collected in Table XI of the SI. We compare three models PET-MAD-1.5-S, PET-SOL-S and PET-SOL-S + D3. We find that the D3 correction is not appropriately reproducing ion–O and O–O RDFs (as already observed for revPBE-D3 functional [75]), which is why in the following results the focus is on the difference between the universal PET-MAD-1.5-S model and the fine-tuned solvation specific counterpart, PET-SOL-S. Figure 2(a) places all ions on a common free-energy scale, plotting ΔF against the position of the PMF minimum, r_min. Be^{2+} is not shown, because no water is found at the first RDF minimum within 5 ns ($g_\mathrm{min} = 0$) and ΔF is not finite; for PET-MAD-1.5-S the same holds for Mg^{2+}. ΔF spans more than an order of magnitude, from below $1\,k_\mathrm{B}T$ for the heavy halides to 6.8–$9.3\,k_\mathrm{B}T$ for the lanthanides and $\approx 12\,k_\mathrm{B}T$ for Mg^{2+}, and decreases with increasing r_max within each group, following the decreasing charge density of the ion. This ordering reproduces the classical distinction between strongly (positively) hydrated ions, with first-shell water exchanges more slowly than in bulk water, and weakly (negatively) hydrated ions such as Rb^+, Cs^+ or I^-, whose shell water exchanges about as fast as bulk water [76]. The two models agree closely on both coordinates: the median absolute difference in ΔF is $0.18\,k_\mathrm{B}T$, and the PMF minima coincide to within a few hundredths of an Å for all ions except Fr^+ and At^-. Nonetheless, uncertainty quantification via Last-Layer-Prediction-Rigidity (LLPR) shows that PET-SOL-S is much more confident in the PMF than PET-MAD-1.5-S. This is especially relevant in weakly associating larger ions such as Cs^+, taken as a representative example, shown in Fig. 1. Figure 2(b) relates PET-MAD-1.5-S committee uncertainties in ΔF and the absolute PET-MAD-1.5-S vs PET-SOL-S errors $\Delta\Delta F$. 63% are within 1σ, 97% within 2σ and all ions within 3σ. The results are roughly Gaussian distributed, suggesting that the uncertainty provided by the LLPR committee is a good measure for assessing how far the model is from a ground truth, which is approximated by the fine-tuned model. Because ΔF is an equilibrium quantity, it is available from the RDF alone also where water exchange is too slow to sample the MRT directly. Figure 2(c) shows that it nonetheless encodes the exchange dynamics: for PET-SOL-S, ΔF depends linearly on $\ln\langle\tau\rangle$ across all 31 ions with a finite barrier ($R^2 = 0.99$). This linear relationship, strengthens the case of approximating the MRT

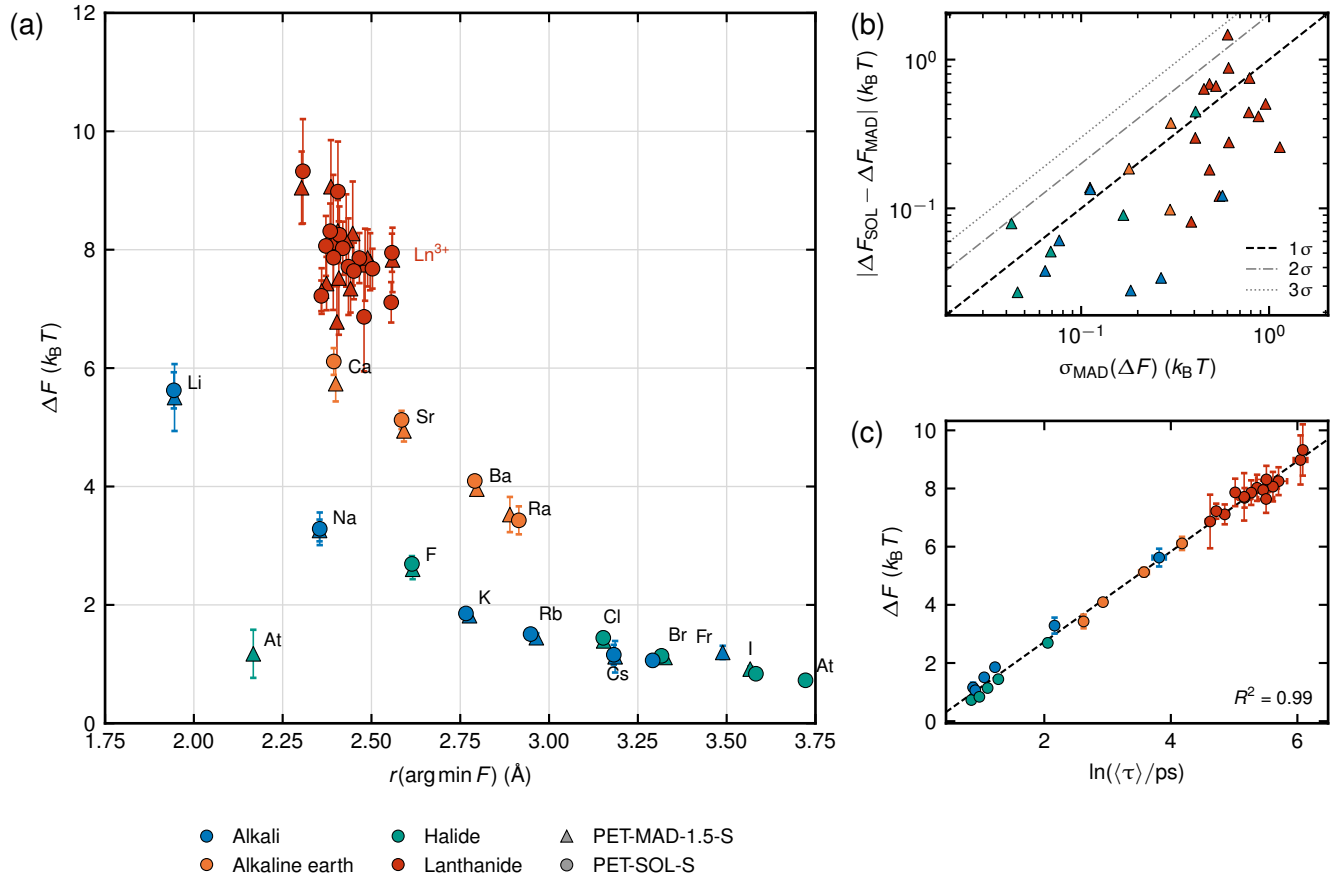


FIG. 2. (a) First-shell free energy difference ΔF against the position of the PMF minimum (first RDF peak), $r_{\max} = \arg\min F$, for every ion in the dilute limit from 5 ns NVT trajectories, comparing PET-MAD-1.5-S (triangles) and PET-SOL-S (circles) on the same axes. (b) Absolute model difference $|\Delta F_{\mathrm{SOL}} - \Delta F_{\mathrm{MAD}}|$ between PET-SOL-S and PET-MAD-1.5-S against the LLPR committee uncertainty $\sigma_{\mathrm{MAD}}(\Delta F)$ of PET-MAD-1.5-S for each ion (log-log); the lines mark 1, 2 and 3σ. (c) ΔF vs the logarithm of mean residence time τ across the ion set for PET-SOL-S, with error bars as defined in Sec. II E 3.

with a static property that can be obtained from the $g(r)$ that can be sampled much more efficiently.

From classical rate theory we can relate the exchange rate to the free energy barrier

$$k_{ex} = A_0 e^{-\beta \Delta F}, \tag{16}$$

$$\ln\langle\tau\rangle = -\ln A_0 + \beta \Delta F. \tag{17}$$

We find a slope of $1.3\, k_B T$. This suggests, that the radial component alone is not sufficient for describing the transition path. Further, the slope is highly dependent on the definition of the cutoff chosen when defining the survival function. Using the r_{min} as cutoff reveals a slope of $1.37\, k_B T$, showing its strong dependence on the choice of cutoff. **TODO:** Perhaps interesting to relate the intercept value to the diffusion of water.

Within the series, both models reproduce the increase of the first-shell radius r_{max}, $\mathrm{Li}^+ < \mathrm{Na}^+ < \mathrm{K}^+ < \mathrm{Rb}^+$ [77], and the corresponding decrease of the residence times, $\mathrm{Li}^+ > \mathrm{Na}^+ > \mathrm{K}^+ > \mathrm{Rb}^+$ [76]. The two models differ for the heaviest alkali ion Fr^+. PET-SOL-S continues the alkali series smoothly: Fr^+ has its first shell 0.11 Å further out than Cs^+ and a slightly lower barrier (1.06 vs. $1.16\, k_B T$), but a slightly longer residence time (4.61 ± 0.08 vs. 4.28 ± 0.06 ps). A residence time of Fr^+ longer than that of Cs^+ was also found in ab initio MD, where it was attributed to non-equilibrium solvent effects rather than to a higher barrier [78]. PET-MAD-1.5-S places the Fr^+ shell 0.30 Å further out than Cs^+, with a higher barrier (1.20 vs. $1.12\, k_B T$) and a residence time of 6.4 ps.

For the alkaline-earth ions, both models reproduce the increase of r_{max}, $\mathrm{Be}^{2+} < \mathrm{Mg}^{2+} < \mathrm{Ca}^{2+} < \mathrm{Sr}^{2+}$ [79], and the decrease of the MRT, $\mathrm{Be}^{2+} > \mathrm{Mg}^{2+} > \mathrm{Ca}^{2+} > \mathrm{Sr}^{2+}$ [76]. For the halides, r_{max} increases as $\mathrm{F}^- < \mathrm{Cl}^- < \mathrm{Br}^- < \mathrm{I}^-$ and the MRT decreases as $\mathrm{F}^- > \mathrm{Cl}^- > \mathrm{Br}^- > \mathrm{I}^-$, reproducing the trend in r_{max} from experiment [80] and the MRT trend from simulations with higher-level many body potentials [15]. The biggest difference between models is observed in this series, as PET-

MAD-1.5-S predicts a strong At^--O bond, corresponding to an effectively uncharged At. Inclusion of solvated At in the MAD-SOL dataset, leads to a correct structural trend among the halides, and a linear relationship between ΔF and $\ln\langle\tau\rangle$.

Both models capture the lanthanide contraction. The first-shell peak moves from 2.56 Å for La^{3+} to 2.30 Å for Lu^{3+}, with larger steps at the beginning of the series, in agreement with the Ln–O distances from EXAFS [81], and the coordination number decreases gradually from 8.9 to 7.9. The decrease is not strictly monotonic. Dy^{3+} lies ≈ 0.015 Å further out than Tb^{3+}, which is close to the resolution of our analysis, and Yb^{3+} lies ≈ 0.1 Å further out than expected from the series, in both models. Since Yb^{3+} is the lanthanide closest to a closed $4f$ shell, this might reflect a partially divalent description of Yb in the reference calculations. In contrast to the structure, the first-shell dynamics and ΔF show no clear trend along the series. ΔF varies between 6.9 and 9.3 $k_\mathrm{B}T$ without a systematic dependence on the atomic number, and the differences between neighbouring ions are comparable to the committee uncertainty. Representative individual RDFs, and within-group ΔF vs $\ln\langle\tau\rangle$ plots for PET-MAD-1.5-S, PET-SOL-S and PET-SOL-S + D3 are collected in the SI V D.

B. Finite-concentration electrolyte solutions

$CaCl_2$, NaCl and CsCl were simulated at finite concentration (Sec. II D 3) with PET-MAD-1.5-S, PET-SOL-S and PET-SOL-S + D3 to probe ion–ion correlations, thermodynamics and dynamics that the single-ion boxes cannot access. The three salts span the lyotropic series from the strongly hydrated kosmotrope Ca^{2+} over Na^+ to the chaotropic Cs^+. The chloride salts are simulated at ≈ 1 M, and box sizes of roughly 25 Å with the exact number of ions and water molecules in each box listed in Table XII of the SI. Table VIII collects the self-diffusion coefficients, the thermodynamic factor Γ, the first-shell cation–O structure and the water reorientation around the cation; Fig. 3 compares the cation diffusivity relative to water (a), Γ (b), the first-shell cation–O distance (c), the contact ion pairing (d) and the water reorientation in the cation hydration shell (e) between salts and against experiment.

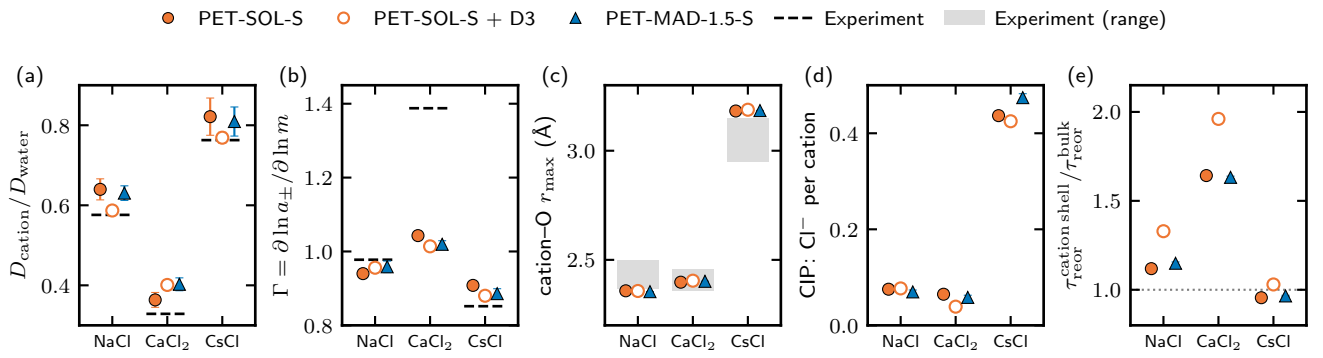
a. Hydration structure The first-shell structure of the cations is essentially unchanged from the dilute limit. The first cation–O peak (Table VIII, Fig. 3c) lies at 2.35–2.36 Å for Na^+, 2.40 Å for Ca^{2+} and 3.18–3.19 Å for Cs^+ in all three models, within 0.01 Å of the single-ion simulations (Table XI). Compared with the experimental range, the Ca^{2+} distance lies within it and Cs^+ lies 0.03–0.04 Å above its upper end. For Na^+, earlier diffraction studies report 2.41–2.50 Å [83], but a recent X-ray diffraction study with higher spatial resolution ($Q_\mathrm{max} = 24$ Å^{-1}), confirmed by Na K-edge EXAFS, resolves the Na–O peak from the O–O peak of water and finds a shorter distance of 2.37–2.38 Å [90], in line with our 2.35–2.36 Å. Experimental hydration numbers scatter widely between methods and concentrations [83], from 4 to 8 for Na^+, 5.5 to 10 for Ca^{2+} and 6 to 8 for Cs^+. The coordination numbers of Na^+ (5.4–5.7) and Ca^{2+} (7.0–7.3) lie within these ranges. For Cs^+ (9.9–11.6) they lie above, but the first minimum of the cation–O RDF is shallow and broad, so that the coordination number changes by more than one water molecule when the cutoff is shifted by 0.2 Å and is not a reliable comparison. Contact ion pairs (Fig. 3d) are rare in the solutions of the strongly hydrated cations, with 0.04–0.07 (Ca^{2+}) and 0.07–0.08 (Na^+) Cl^- per cation, but frequent for Cs^+ (0.43–0.47): Ca^{2+} and Na^+ retain their hydration shells, whereas the weakly hydrated Cs^+ and Cl^- readily pair. The three models agree closely; the D3 correction slightly increases the coordination numbers of all cations and reduces the number of Ca^{2+}–Cl^- contact pairs.

b. Diffusivity (**CMnote:** Mention that diffusivity was not possible in the single IO systematic study due to convergency issue) In order to assess the models' capabilities in reproducing transport properties of ionic solutions, we compare self-diffusivities of ions and water molecules. All three models simulated at 330 K systematically overestimate ionic self-diffusion with respect to the 298 K references. Water diffusion D_O is fastest in every system but is itself overestimated by 31–50 %, so that a large part of the bias originates in the water network rather than in ion-specific solvation. To account for the thermal effect, we normalise ion diffusivity with self diffusivity of water fro mthe same simulations. The *relative* cation/water mobility tracks hydration strength correctly across all three models: $D_\mathrm{cation}/D_{\mathrm{H_2O}}$ is smallest for kosmotropic Ca^{2+}, intermediate for Na^+ and close to unity for chaotropic Cs^+. Clearly, lyotropic tendencies are well captured, as the ion mobility follows the hydration strength. This is an expected result as a strong hydration shell is expected to slow down ion movement. All models predict self diffusivity coefficients well, and are close to experiment. Model differences are small, with no clear winner.

c. Hydration shell dynamics A good estimator for structure-making or structure breaking is the autocorrelation of water orientation around the ions. Relative to bulk water the reorientation is slowed down in the first solvation shell for Ca^{2+} by a factor of 1.6-2.0 and a factor of 1.1-1.3 for Na^+, while water reorients roughly at the same rate around Cs^+. In bulk water, the slow time constant of the bi-exponential fit, $\tau_2 = 1.8$–2.5 ps, is close to the experimental reorientation time of 2.5 ps at room temperature [91]. The three models agree on the ordering, with the dispersion correction strengthening the slowdown for Ca^{2+} and Na^+. Further, we compared water–water hydrogen bond dynamics. We find that hydrogen bonds are more stable in the solvation shell of Ca^{2+}, while there is not a strong influence of Na^+ and Cs^+ on hydrogen bond lifetimes, relative to the bulk. Results are shown in the SI XIII.

TABLE VIII. Transport, thermodynamic and structural properties of the ≈ 1 M finite-concentration electrolyte solutions (Sec. II D 3): self-diffusion coefficients (10^{-5} cm^2 s^{-1}) of the cation, Cl^- and water O from Einstein-relation MSD fits (Sec. II E 4), Yeh-Hummer-corrected using the model's own pure-water viscosity; the salt thermodynamic factor $\Gamma \equiv \partial \ln a_\pm / \partial \ln c$ from the u_2-corrected KBI estimator (Sec. II E 6), converted from the simulated molarity scale to the molality scale of the experimental reference; the position r_{max} (Å) of the first cation–O RDF peak and the cation–O coordination number CN to the first minimum (Sec. II E 1); and the water reorientation time in the first hydration shell of the cation relative to bulk water, $\tau_{reor}^{shell}/\tau_{reor}^{bulk}$ (Sec. II E 5). Uncertainties are the combined-replica SEM (D, τ_{reor} ratio) or the SEM over 12 blocks (Γ, r_{max}, CN). Γ_{exp} is the Pitzer-equation value [82] evaluated at each system's own simulated molality (averaged over the three models' slightly different simulated concentrations); experimental r_{max} and CN are the range of the X-ray and neutron diffraction and EXAFS values compiled in Ref. 83.

Salt	Model / reference	D_{cation}	D_{Cl^-}	D_O	Γ	r_{max}	CN	$\tau_{reor}^{shell}/\tau_{reor}^{bulk}$
$CaCl_2$ (Ca^{2+})	PET-MAD-1.5-S	1.14 ± 0.04	2.34 ± 0.07	2.83 ± 0.04	1.019 ± 0.010	$2.40 \pm <0.01$	$7.1 \pm <0.1$	1.63 ± 0.01
	PET-SOL-S	1.01 ± 0.05	2.25 ± 0.05	2.78 ± 0.04	1.043 ± 0.008	$2.40 \pm <0.01$	$7.0 \pm <0.1$	1.64 ± 0.01
	PET-SOL-S + D3	1.05 ± 0.01	2.33 ± 0.08	2.61 ± 0.01	1.014 ± 0.011	$2.40 \pm <0.01$	$7.3 \pm <0.1$	1.96 ± 0.01
	experiment	0.62[a]	1.54[a]	1.88[b]	1.388	2.36–2.46[c]	5.5–10[c]	–
CsCl (Cs^+)	PET-MAD-1.5-S	2.76 ± 0.12	2.75 ± 0.05	3.41 ± 0.03	0.886 ± 0.013	$3.18 \pm <0.01$	9.9 ± 0.1	$0.97 \pm <0.01$
	PET-SOL-S	2.76 ± 0.15	2.83 ± 0.04	3.36 ± 0.05	0.908 ± 0.008	$3.18 \pm <0.01$	10.1 ± 0.1	$0.96 \pm <0.01$
	PET-SOL-S + D3	2.48 ± 0.04	2.72 ± 0.14	3.22 ± 0.02	0.880 ± 0.012	$3.19 \pm <0.01$	$11.6 \pm <0.1$	$1.03 \pm <0.01$
	experiment	1.87[d]	1.75[e]	2.45[b]	0.852	2.95–3.15[c]	6–8[c]	–
NaCl (Na^+)	PET-MAD-1.5-S	2.02 ± 0.05	2.81 ± 0.04	3.21 ± 0.03	0.959 ± 0.016	$2.35 \pm <0.01$	$5.5 \pm <0.1$	1.15 ± 0.01
	PET-SOL-S	2.01 ± 0.08	2.54 ± 0.06	3.14 ± 0.04	0.940 ± 0.012	$2.36 \pm <0.01$	$5.4 \pm <0.1$	1.12 ± 0.01
	PET-SOL-S + D3	1.73 ± 0.03	2.61 ± 0.08	2.94 ± 0.02	0.956 ± 0.011	$2.36 \pm <0.01$	$5.7 \pm <0.1$	1.33 ± 0.01
	experiment	1.25[f]	1.77[g]	2.17[b]	0.978	2.37–2.50[h]	4–8[c]	–



[a] Ref. [84].
[b] Ref. [85].
[c] Ref. [83].
[d] Ref. [86].
[e] Ref. [87].
[f] Ref. [88].
[g] Ref. [89].
[h] Refs. [83, 90].

FIG. 3. Finite-concentration (≈ 1 M) observables against experiment and between salts (values in Table VIII where listed), for PET-SOL-S, PET-SOL-S + D3 (open markers) and PET-MAD-1.5-S; dashed segments are experiment. (a) Cation self-diffusion coefficient relative to that of water in the same simulation (Yeh–Hummer corrected as in Table VIII). (b) Thermodynamic factor Γ on the molality scale against the Pitzer value. (c) Position of the first cation–O RDF peak against several experimental results. (d) Number of Cl^- in contact with a cation (contact ion pairs), integrated to the first minimum of the cation–Cl RDF. (e) Water reorientation time (C_2, amplitude-weighted mean) in the first hydration shell of the cation, defined by the first minimum of the cation–O RDF, relative to bulk water. Error bars are replica (a, e) or block (b–d) standard errors.

The mean residence time of water in the first shell of the cation, computed as in the dilute limit (Table XIII), follows the same hydration-strength ordering, Ca^{2+} (79–100 ps) ≫ Na^+ (13.3–13.9 ps) ≫ Cs^+ (4.2–5.0 ps), and stays close to the dilute-limit values (Table XI), which supports the picture of strongly kosmotropic Ca^{2+} and weakly chaotropic Cs^+.

d. Thermodynamics The thermodynamic factor Γ is a measure of non-ideality of a component. It is tightly linked to the excess volume around the ions, and can be derived from ion–ion and ion–O Kirkwood-Buff integrals. For an ideal solution $\Gamma = 1$. We find, that

NaCl and CsCl solution factors are well predicted compared to experimental reference from Pitzer [82]. However, all models underestimate the correction to $CaCl^{2+}$ drastically. This points to a too strong Ca–Ca interaction in all models, hinting at a problem in the reference data. Fine-tuning on salt specific data could alleviate the deficiency, but is outside the scope of this work.
Averaged across all three salts, PET-SOL-S + D3 gives the lowest Γ error (9.6 %, vs. 10.9 % for PET-MAD-1.5-S and 12.3 % for PET-SOL-S). $CaCl_2$ is the exception, underestimating Γ by 25–29 % in every model.

This indicates that the models generally reproduce the correct ion-ion correlations, and are capable of capturing thermodynamic properties and dynamic trends.

C. Bulk solvent densities and viscosities

Figure 4 and Table IX compare the densities and shear viscosities of 16 molecular liquids, simulated with PET-SOL-S + D3, with experiment. Because r^{2}SCAN recovers only short- and intermediate-range dispersion, the D3 correction [31] is essential for the apolar and weakly polar liquids, whose cohesion is dominated by van der Waals interactions. With it, densities are reproduced with a mean absolute relative error of 3%, or 1.9% if water and the two chlorinated solvents are excluded. The polar aprotic solvents (DMF, DMSO, THF, diethyl ether and the two carbonates) lie within 1.5% of experiment, while the alcohols (3–6%), the aromatics and hexane (2–3%) are slightly too dense. All these deviations are smaller than the LLPR uncertainty of 0.03–0.07 g cm^{-3}, which is therefore a conservative estimate of the model error for these liquids. The exceptions are chloroform and CCl_4, which are 7% and 9% too light. These are the only liquids whose density is underestimated by more than 1%, and CCl_4 also carries the largest LLPR uncertainty of the series.

Water behaves differently. PET-SOL-S without dispersion gives 0.989 g cm^{-3}, within 1% of experiment. Adding D3 makes water 11% too dense, presumably because r^{2}SCAN already binds the hydrogen-bond network strongly and the correction double-counts part of the attraction. PET-MAD-1.5-S shows the same behaviour, so this is a property of the reference level rather than of the fine-tuning. The viscosity of water is much less sensitive to the correction: 0.68 and 0.75 mPa s with and without D3, both 16–24% below experiment.

Viscosity is a far more demanding test than density, as it depends on the time scales of molecular rearrangements, and errors in the intermolecular interactions are amplified exponentially through the activation energy of viscous flow. Across the series the mean absolute relative error is 18%, with no systematic sign. DMF, THF, diethyl ether and toluene agree with experiment within 5%, and ethyl methyl carbonate, DMSO and methanol within 15%. The hydrogen-bonded liquids ethanol, propan-2-ol and acetic acid are about 30% too fluid, and the chlorinated solvents 30–40%, consistent with their underestimated density. Benzene, hexane and ethylene carbonate are 18–31% too viscous. For the four liquids also simulated with PET-MAD-1.5-S + D3, densities change by less than 2% upon fine-tuning, while viscosities differ by up to a factor of 1.5. The parent model is closer to experiment for acetic acid and benzene, PET-SOL-S for chloroform. Fine-tuning on MAD-SOL thus leaves the densities of the neat solvents essentially unchanged, whereas viscosities, a far more sensitive probe of the potential, shift by up to 50%, without a consistent improvement over the parent model with respect to experiment. (**CMnote:** All liquids are simulated at 330 K and compared with room-temperature experiments (see Methods). For the organic solvents a 30 K higher temperature lowers both density and viscosity, so the density overestimate would grow and the viscosity underestimate shrink at 298 K. Should this be stated, or should the organics be compared with experiment at 330 K?)

D. Deep Eutectic solvents

Deep eutectic solvents (DESs) are solvents composed of a binary or ternary mixture of compounds that have a decreased melting temperature compared to the pure isolated components. Generally, DESs are composed of hydrogen bond acceptors and neutral hydrogen bond donors that when mixed together form a complex hydrogen bonding network with interactions between neutral and charged species. As opposed to traditional organic solvents, DESs are highly biodegradable, nonflammable, and have low vapor pressures, making them more environmentally friendly [92]. Additionally, mixtures formed with DESs and water allow for tunable static and dynamic properties [93].

Due to the complex intermolecular interactions present in DESs, their simulation with molecular dynamics is quite difficult. When fine tuned on DES experimental data, classical force fields like OPLS-DES can often reproduce static properties of DESs quite well (e.g. density) [92, 93]. However, for dynamic properties like viscosity, these classical force fields can often fail by several orders of magnitude. The strong intermolecular interactions present in DESs lead to long structural relaxation times, requiring longer simulation times to converge the viscosity due to long-lived stress correlations. Additionally, viscosity is much more sensitive to the potential being used than density because of the dependence of viscosity on intermolecular forces and the timescales of molecular rearrangements that are governed by the underlying potential. Scaling charges and adjusting Lennard-Jones parameters in classical force fields can help improve agreement with experiment, but no univer-

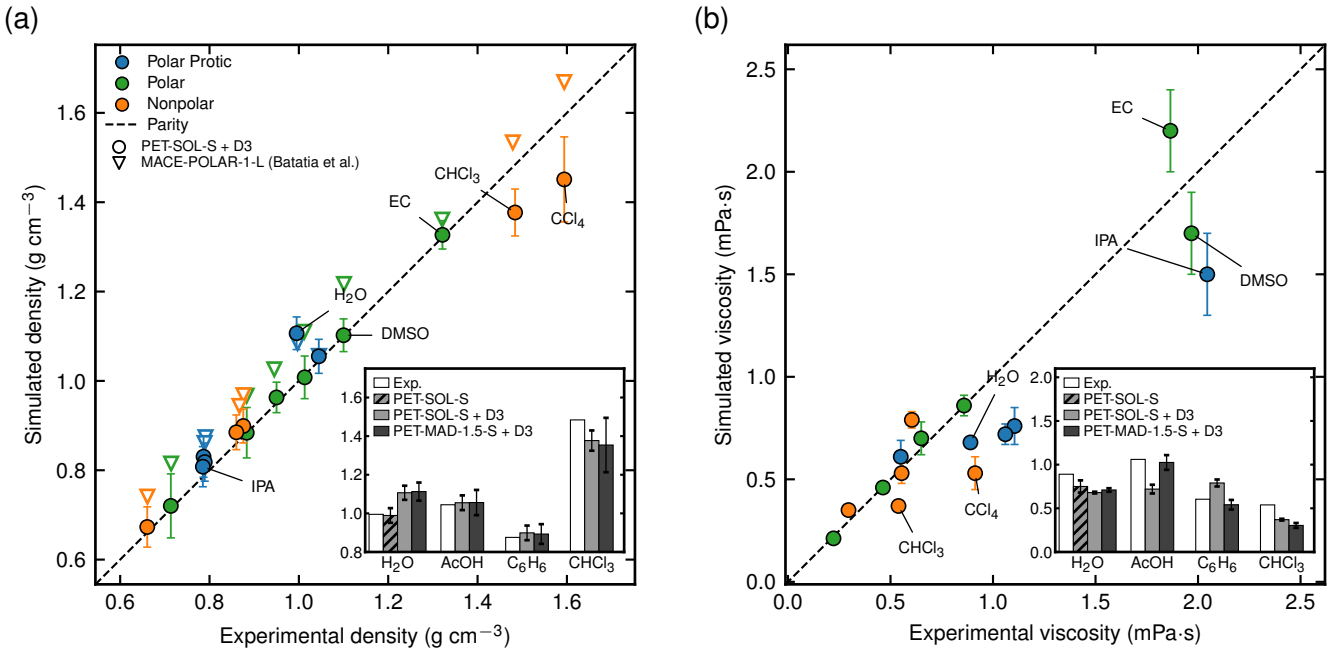


FIG. 4. Parity plots of solvent properties densities (a) and viscosities (b) obtained from PET-SOL-S + D3. Insets show results from multiple models for comparison. Densities were obtained from bulk NPT simulations, while viscosities were obtained from stress autocorrelation in NVT simulations. Density errorbars show model uncertainty obtained from LLPR. Viscosity errors are obtained from cepstral analysis.

TABLE IX. (**CMnote:** Remove stat uncertainty from density, swap with LLPR, discuss that the stat unce is << LLPR)Simulated and experimental liquid densities and viscosities. All models are PET-SOL-S + D3 unless otherwise specified. Uncertainties of the simulated values are block standard errors; σ_{LLPR} is the LLPR committee uncertainty of the density, obtained by reweighting the NPT trajectory onto the committee members with the linear cumulant expansion of Ref. 56. D3 is not included in the committee and does not contribute to σ_{LLPR}.

Solvent (model)	ρ (g cm^{-3})	σ_{LLPR} (g cm^{-3})	Expt. ρ (g cm^{-3})	η (mPa · s)	Expt. η (mPa · s)
Water	1.1067 ± 0.0003	0.04	0.995	0.68 ± 0.01	0.89
Water (PET-SOL-S)	0.9891 ± 0.0004	0.04	0.995	0.75 ± 0.07	0.89
Water (PET-MAD-1.5-S)	0.9982 ± 0.0003	–	0.995	0.70 ± 0.05	0.89
Water (PET-MAD-1.5-S + D3)	1.1128 ± 0.0003	0.05	0.995	0.71 ± 0.02	0.89
Methanol	0.8304 ± 0.0008	0.05	0.7866	0.61 ± 0.08	0.55
Ethanol	0.8183 ± 0.0007	0.04	0.7893	0.76 ± 0.09	1.105
Propan-2-ol	0.808 ± 0.001	0.04	0.78509	1.5 ± 0.2	2.045
Acetic acid	1.055 ± 0.001	0.04	1.0446	0.72 ± 0.05	1.06
Acetic acid (PET-MAD-1.5-S + D3)	1.056 ± 0.001	0.07	1.0446	1.03 ± 0.08	1.06
N,N-dimethylformamide	0.963 ± 0.001	0.03	0.95	0.86 ± 0.05	0.8586
Dimethyl sulfoxide	1.1022 ± 0.0006	0.04	1.1000	1.7 ± 0.2	1.968
Tetrahydrofuran	0.884 ± 0.002	0.06	0.8833	0.46 ± 0.02	0.4631
Diethyl ether	0.7202 ± 0.0009	0.07	0.7134	0.212 ± 0.005	0.222
Ethylene carbonate	1.327 ± 0.001	0.03	1.3214[a]	2.2 ± 0.2	1.8653[a]
Ethyl methyl carbonate	1.008 ± 0.001	0.05	1.013	0.70 ± 0.08	0.65
Benzene	0.899 ± 0.002	0.04	0.8756	0.79 ± 0.04	0.604
Benzene (PET-MAD-1.5-S + D3)	0.893 ± 0.001	0.05	0.8756	0.54 ± 0.06	0.604
Toluene	0.885 ± 0.001	0.04	0.86	0.53 ± 0.05	0.5542
Hexane	0.673 ± 0.001	0.05	0.6606	0.35 ± 0.02	0.2952
Chloroform	1.377 ± 0.002	0.05	1.484	0.37 ± 0.01	0.539
Chloroform (PET-MAD-1.5-S + D3)	1.354 ± 0.003	0.14	1.484	0.30 ± 0.03	0.539
Tetrachloromethane	1.451 ± 0.003	0.10	1.594	0.53 ± 0.08	0.91284

[a] Expt. at 313 K.

sal schema exists for all DES systems [92]. Consequently, the accurate simulation of DESs presents a challenging test case for evaluating the capabilities of PET-SOL on systems with strong intermolecular interactions and slow dynamics that are far outside of PET-SOL's training set. Here, we evaluate PET-SOL-S + D3 and PET-MAD-1.5-S + D3 on two DES systems, reline and ethaline mixed with various molar fractions of water, and compare their predictions of density and viscosity against experimental measurements.

Reline and ethaline are both 1:2 mixtures of choline chloride with the hydrogen bond donors urea and ethylene glycol, respectively. Importantly, reline exhibits significant urea-urea hydrogen bonding. The strong urea-urea and urea-Cl- intermolecular interactions present in reline result in a substantially higher viscosity of reline compared to ethaline at lower molar fractions of water [94]. Therefore, reline mixtures serve as an even harder test case for our models than ethaline mixtures given their strong intermolecular interactions and slow dynamics.

As shown in Fig. 5, both PET-SOL-S + D3 and PET-MAD-1.5-S + D3 reproduce experimental densities of reline and ethaline fairly well across the range of molar fractions of water chosen, with deviations under 8% for all compositions. For both models and both DESs, however, as the molar fraction of water increased, the discrepancy between density obtained via simulation and experimental density also increased. This can be attributed to the D3 wrapper of the models overcorrecting for the attractive dispersion intermolecular forces in more hydrous environments - this behavior was observed previously in relation to water having a higher density for D3 models.

A much more difficult test of these models' capabilities is obtaining the viscosity for these DES systems. These results are shown in figure 5. For lower molar fractions of water where DESs have higher viscosities, cepstral analysis yields larger uncertainties in estimated viscosities because of longer relaxation times that cause stress correlations to converge more slowly. While our estimated viscosities at these lower mole fractions of water have higher absolute differences in relation to the experimental viscosity than at less viscous higher water mole fraction mixtures, they remain within the correct order of magnitude. Additionally, our PET-SOL-S + D3 model outperforms the PET-MAD-1.5-S + D3 model in estimating neat (xw=0) viscosities of reline and ethaline. For neat ethaline, the experimental viscosity falls within the uncertainty of the PET-SOL-S + D3 estimated viscosity. Longer simulation times may have helped to reduce the uncertainties associated with these more slowly relaxing systems, though systematic errors related to the models would remain. Importantly, though, our results showcase that PET-SOL-S + D3 can predict viscosities of strongly interacting DESs within reasonable accuracy despite these systems lying far outside of its training distribution and without any model specific fine tuning. For comparison, Fig. 5 also shows the classical GAFF simulations with scaled charges of Celebi et al. [94] at 303 K for both DESs. For the neat DESs, GAFF gives 39 ± 4 mPa s for ethaline and 511 ± 25 mPa s for reline, compared with 38 ± 11 and 494 ± 271 mPa s for PET-SOL-S + D3 and 17 ± 2 and 153 ± 50 mPa s for PET-MAD-1.5-S + D3, so that PET-SOL-S + D3 is comparable to a classical force field parametrised for these liquids. The comparison is only approximate, since experiment, GAFF and the MLIP simulations refer to 298, 303 and 330 K, respectively.

E. Water autoionization

The solvent ionization, as the simplest chemical process happening in the solvent, only involves the solvent molecules themselves. This simple process, however, is still challenging to foundation MLIPs, because it demands a good description of the bond cleavaging and forming process from the MLIPs, while these chemical spaces are usually poorly covered by existing datasets. Here, we take water as an example. By applying the same method used in [52], we computed the pK_w with PET-MAD-1.5-S, PET-SOL-S, PET-SOL-S + D3, and fine-tuned versions of PET-SOL-S, as shown in figure 6.

The classical values of pK_w given by PET-MAD-1.5-S and PET-SOL-S are close, 11.61 and 11.82, respectively. After nuclear quantum effects (NQEs) correction are added to them, pK_w becomes 8.30 and 7.73, which substantially underestimate the experiment value of 13.40 at 330 K, consistent with the conclusion of Ref. [96] on SCAN functional. Adding D3 correction to PET-SOL lowered the classical value (10.60), but did not show significant influence on the quantum-corrected value (7.89). To test whether the underestimation is originated from a poor coverage of the transition state of the training dataset, we relabeled the bulk water dataset of Ref. [51] at the level of theory of this work, and fine-tuned PET-SOL on it. However, the improvement is marginal, where the classical value remains almost unchanged (11.90) and the quantum corrected value becomes 8.36. These results, along with previous study [96], suggest that r^{2}SCAN might not be the most ideal functional to study the water autoionization.

Given the good performance of revPBE0-D3 on water [51] and acid-base equilibria [52], we fine-tuned PET-SOL on the original dataset of Ref. [51]. The results are greatly improved. The classical value becomes 16.70, and the quantum corrected value is 13.12, close to the experiment value. However, both values are lower then the values reported in Ref. [52], which might be attributed to the fact that the dataset of Ref. [51] is not crafted for the purpose of studying water autoionization, and thus the coverage of the water dissociation region is slightly worse than Ref. [52], leading to the discrepancy of the pK_w values between these two studies.

Nonetheless, current results demonstrate that our foundation models, PET-MAD and PET-SOL, are able

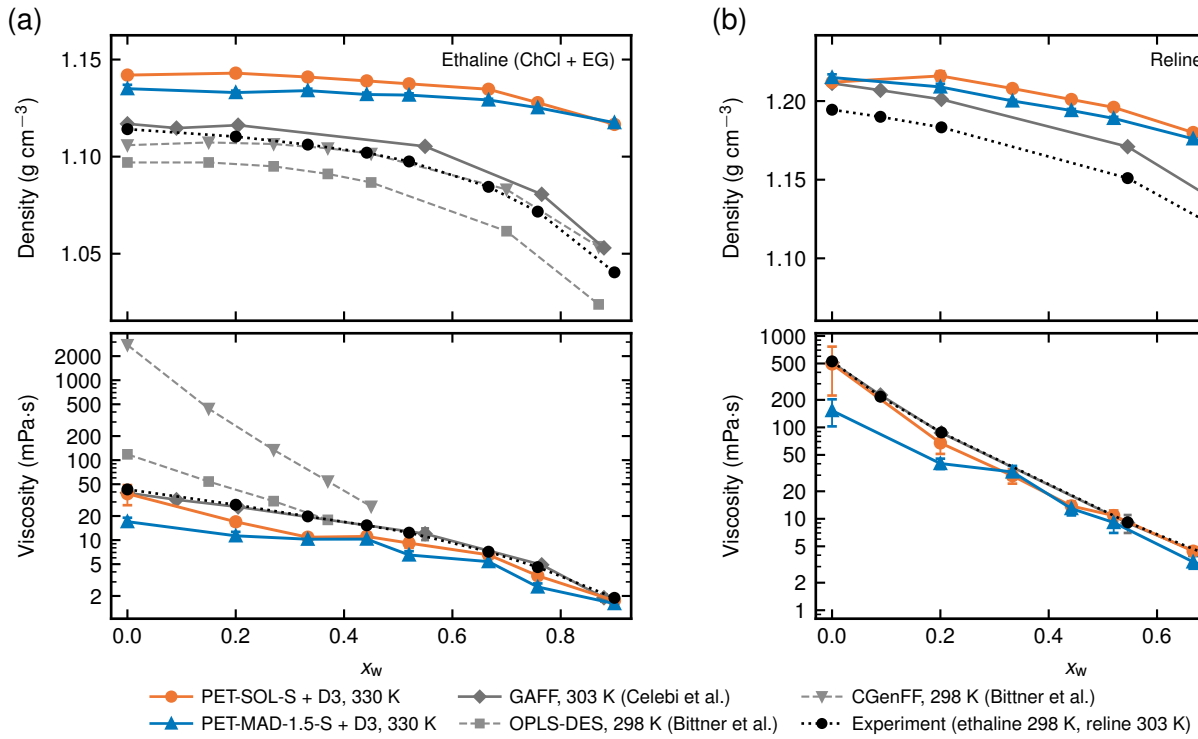


FIG. 5. Densities (top) and viscosities (bottom, logarithmic scale) of (a) ethaline and (b) reline as a function of the water mole fraction x_w, simulated with PET-SOL-S + D3 and PET-MAD-1.5-S + D3 at 330 K and compared with experiment (dotted) at 298 K for ethaline and at 303 K for reline (Ref. 95), the classical GAFF simulations with scaled charges of Ref. 94 at 303 K, and OPLS-DES and CGenFF for aqueous ethaline at 298 K (dashed, digitised from Ref. 93). Error bars are block standard errors (density) and cepstral-analysis uncertainties (viscosity).

to describe the reaction of water autoionization and sample the related chemical space. Though the results deviate from the experiment value, the foundation models can serve as a chemical space explorer and help build the dataset for fine-tuning efficiently. The fine-tuning dataset can be labeled with the same level of theory as the one of the foundation model, if one wants to improve the coverage of the chemical space, or labeled with another level of theory that is more suitable for the chemical process of interest. These things together indicate that the door toward an *ab initio* quantitative modeling of solvent chemistry has been opened by our models.

IV. CONCLUSIONS

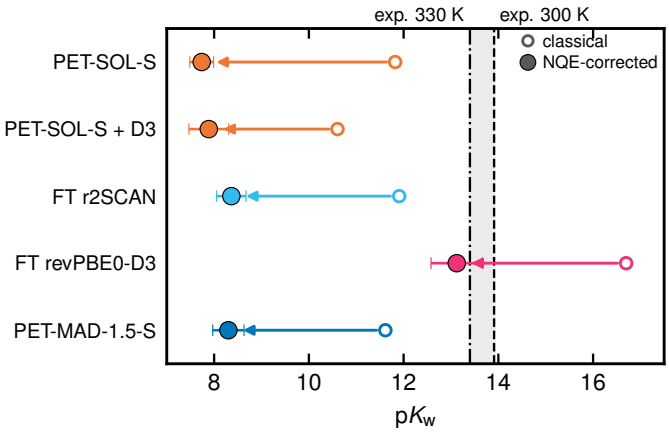


FIG. 6. The water autoionization constants, presented in pK_w.

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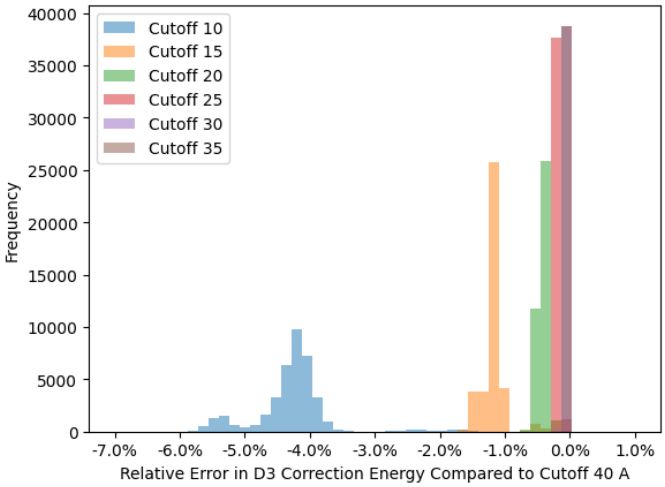


FIG. 7. The convergence test of the cutoff radius of DFTD3 correction. The unit of the tested cutoffs is in angstrom.

V. SI

A. D3 Wrapper

DFT-D3 correction [31] with the BJ damping [97] is employed. The correction energy writes the following:

$$E = -\frac{1}{2}\sum_{A\neq B}\sum_{n=6,8} s_n \frac{C_n^{AB}}{r_{AB}^n + f(R_0^{AB})^n}, \tag{18}$$

where A and B denote atoms, s_n is a functional-dependent parameter, C_n^{AB} represents the n-th order dispersion coefficients of atom pair AB, r_{AB} is the distance between the AB atom pair, with

$$R_0^{AB} = \sqrt{\frac{C_8^{AB}}{C_6^{AB}}}, \tag{19}$$

and

$$f(R_0^{AB}) = a_1 R_0^{AB} + a_2, \tag{20}$$

where a_1 and a_2 are also functional-dependent parameters.

To ensure a seamless integration of DFT-D3 correction and our models, the former is reimplemented in the `metatomic` package [33]. DFTD3 is implemented as a *wrapper*, which does not modify the existing models; instead, it takes the calculated energy from the wrapped model, calculates the correction based on the input configuration, and adds the correction dispersion energy to the energy output. The correction of the forces is then automatically calculated through auto-differentiation.

To select the optimal cutoff radius, we scanned different cutoff radii of DFTD3 correction, as shown in 7. When the radius is set as 20 Å, the correction can recover 99.5 % of the correction of the one calculated with a 40 Å cutoff, while much faster than it. Therefore, we choose 20 Å as the cutoff radius of the correction of the S model, and choose 15 Å as the cutoff radius of the correction of the XS model.

The functional-dependent parameters are chosen to be the same as the parameters optimized for r^{2}SCAN, where $a_1 = 0.49484001, s_8 = 0.78981345, a_2 = 5.73083694, s_6 = 1.0$ [98].

B. Parity plots for the released models

This section collects the correlation between predicted and reference quantities underlying the error metrics quoted in the main text. Five models are shown, in the same order throughout: the two released models, PET-SOL-S and PET-SOL-XS; PET-MAD-1.5-S-40nn, the control obtained by applying the identical recipe to MAD-1.5 alone; and

TABLE X. Inference speeds (ns/day) of the released models for water systems of varying size: NVT MD with LAMMPS and the `metatomic` pair style (Kokkos) on a single NVIDIA H100 GPU, timed over 1 000 steps after 1 000 warm-up steps. D3 cutoffs: 20 Å (S) and 15 Å (XS). –: exceeds the GPU memory.

Model	System size (atoms)		
	300	3000	30 000
XS	3.329	0.801	0.087
XS + D3	2.572	0.607	0.066
S	1.599	0.307	–
S + D3	1.135	0.232	–

the two unmodified foundation models, PET-MAD-1.5-S and PET-MAD-1.5-XS. Each is evaluated on the two held-out test sets, 3892 MAD-SOL configurations (Fig. 8) and 18 314 MAD-1.5 configurations (Fig. 9), neither of which was seen during training or model selection.

Energies are atomization energies reported per atom throughout; forces are given per Cartesian component, pooled over every atom of every configuration in the set. Each figure has three rows. The first is the energy parity plot proper. On its own it is close to uninformative here, because the per-atom energies span several eV across the compositions present while the errors are of order meV: every model lies on the diagonal. The second row therefore shows the distribution of the residual, predicted minus reference, as a histogram; every model shares the same bin edges and the same axis ranges, so the distributions can be compared bin by bin, and this is the panel that actually discriminates the models. Its range spans the 99th percentile of the residual taken over all five models, so about one configuration in a hundred falls outside the frame. The third row is the force parity plot; its axes are clipped to the 99.9th percentile of the force magnitude over all five models, since a handful of components reach tens of eV Å^{-1} and would otherwise compress the whole distribution onto the diagonal. Quoted MAEs are computed over all points, including any omitted from the scatter for legibility: the force panels are drawn from a random subsample of at most 1.5×10^5 components, while the energy panels show every configuration.

On MAD-SOL (Fig. 8) the residuals show that a large part of the foundation models' error is a systematic offset rather than scatter: they underbind by -8.0 meV atom^{-1} on average in the S size and by -15.0 meV atom^{-1} in the XS size, whereas the released models are essentially unbiased (-0.08 and -0.25 meV atom^{-1}). The offset is not the whole story: removing it leaves the foundation models at 6.0 and 8.2 meV atom^{-1} MAE, still four to six times the 1.4 and 3.5 meV atom^{-1} of the released models, so fine-tuning on MAD-SOL corrects both a shift of the energy scale and a genuine loss of accuracy. PET-MAD-1.5-S-40nn sits between the two, with a residual offset of -5.1 meV atom^{-1}, which shows that neither the change in neighbour count nor the extra optimisation accounts for the improvement on its own.

The configurations separate into two groups along the abscissa. The sparse group above -5000 meV atom^{-1} is made up almost entirely of the two chlorinated solvents, chloroform and tetrachloromethane (363 of its 369 configurations); their smaller cohesive energy per atom reflects the heavy halogen content. The dense group below that value contains the remaining fifteen solvents. Both groups behave the same way from model to model, so the offset discussed above is a property of the model rather than of a particular solvent class.

On MAD-1.5 (Fig. 9) every model is essentially unbiased (mean signed errors of at most 2 meV atom^{-1}) and the residuals are symmetric with heavy tails, a few hundred configurations lying several hundred meV atom^{-1} from the diagonal. Those tails are what makes the RMSE quoted in the main text exceed the MAE by a factor of four on this set while the two are comparable on MAD-SOL. Comparing the first column with the third and fourth confirms the point made in the main text: adding MAD-SOL to the training pool leaves performance on the parent chemical space unchanged. Note that the force axes span a far wider range here than on MAD-SOL, since MAD-1.5 contains strongly compressed and rattled configurations with force components an order of magnitude larger than anything in a liquid.

C. Model performance

Models are released in four variants: S, S+D3, XS, and XS+D3, with inference speeds reported in Table X for water systems ranging from 300 to 30 000 atoms. To benchmark model performance, simulations were run in LAMMPS [48] in the NVT ensemble at experimental density of water. For 3 000 atoms the XS model reaches 0.801 ns/day, roughly 2.6× faster than S (0.307 ns/day), with the performance gap widening at larger system sizes. Only XS scales to 30 000-atom systems at practical speeds (0.087 ns/day), making it the model of choice for large-scale or exploratory simulations. The D3 correction adds an overhead of ∼32% for both S and XS at 3 000 atoms (29% for XS and 41% for

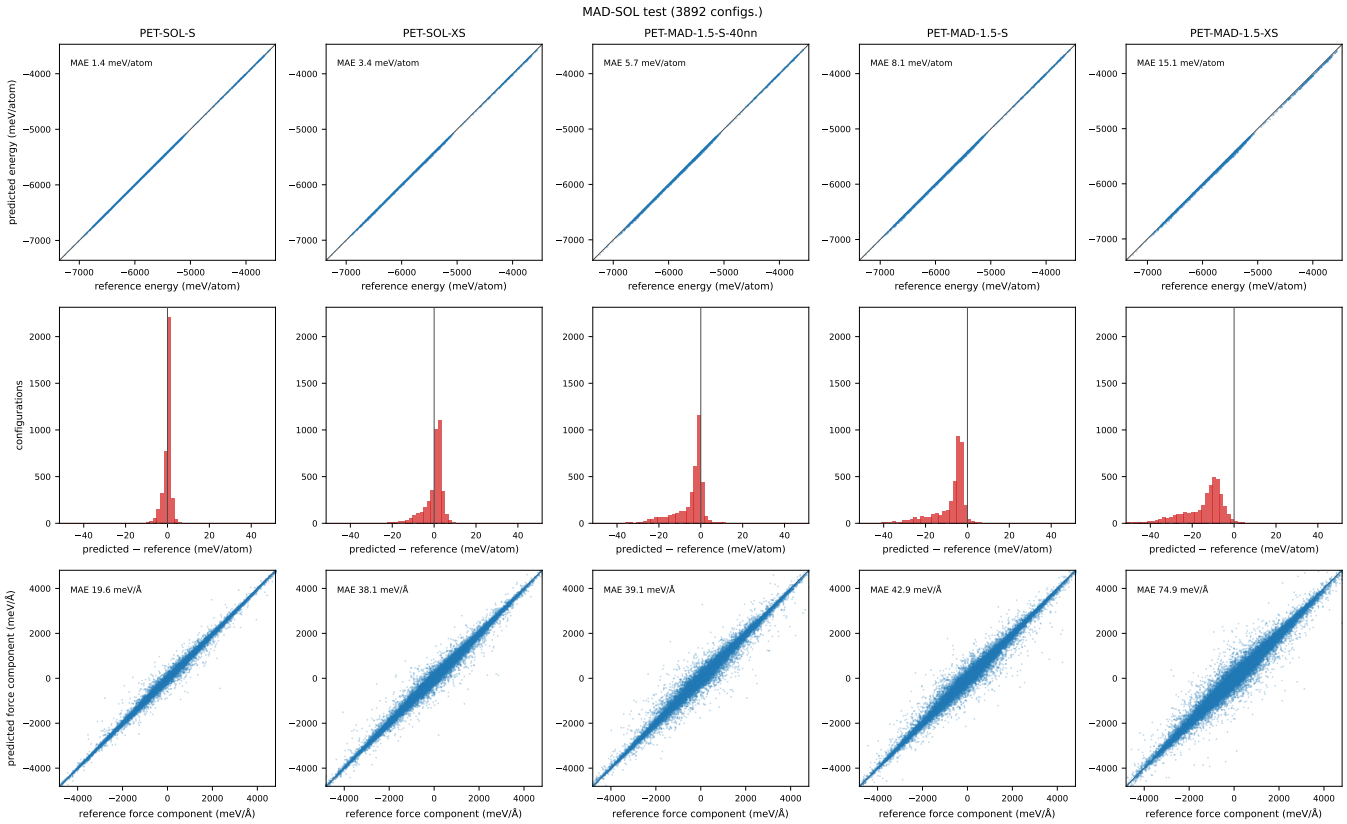


FIG. 8. Parity plots on the MAD-SOL test set (3892 configurations). Top row: atomization energy per atom. Middle row: histogram of the residual, predicted minus reference. Bottom row: Cartesian force components. Columns, from left to right: PET-SOL-S, PET-SOL-XS, PET-MAD-1.5-S-40nn, PET-MAD-1.5-S and PET-MAD-1.5-XS.

S at 300 atoms). For production simulations at moderate system sizes, PET-SOL-S + D3 provides the best accuracy, while PET-SOL-XS + D3 is recommended for large-scale sampling where throughput is the priority.

D. Mean residence time vs. RDF structure across the ion scan

Figure 10 gives the per-category counterpart of Fig. 2(c): the first-shell free-energy barrier ΔF (Sec. II E 1) against the logarithm of the ion–water mean residence time $\langle\tau\rangle$ (Sec. II E 3), one row per ion category (alkali, alkaline-earth, halide, lanthanide) and one column per released potential: the PET-MAD-1.5-S foundation model, PET-SOL-S, and PET-SOL-S with the D3 dispersion correction (Sec. V A). Ions are individually labeled here, since each panel covers only the handful of ions in one category rather than the full scan.

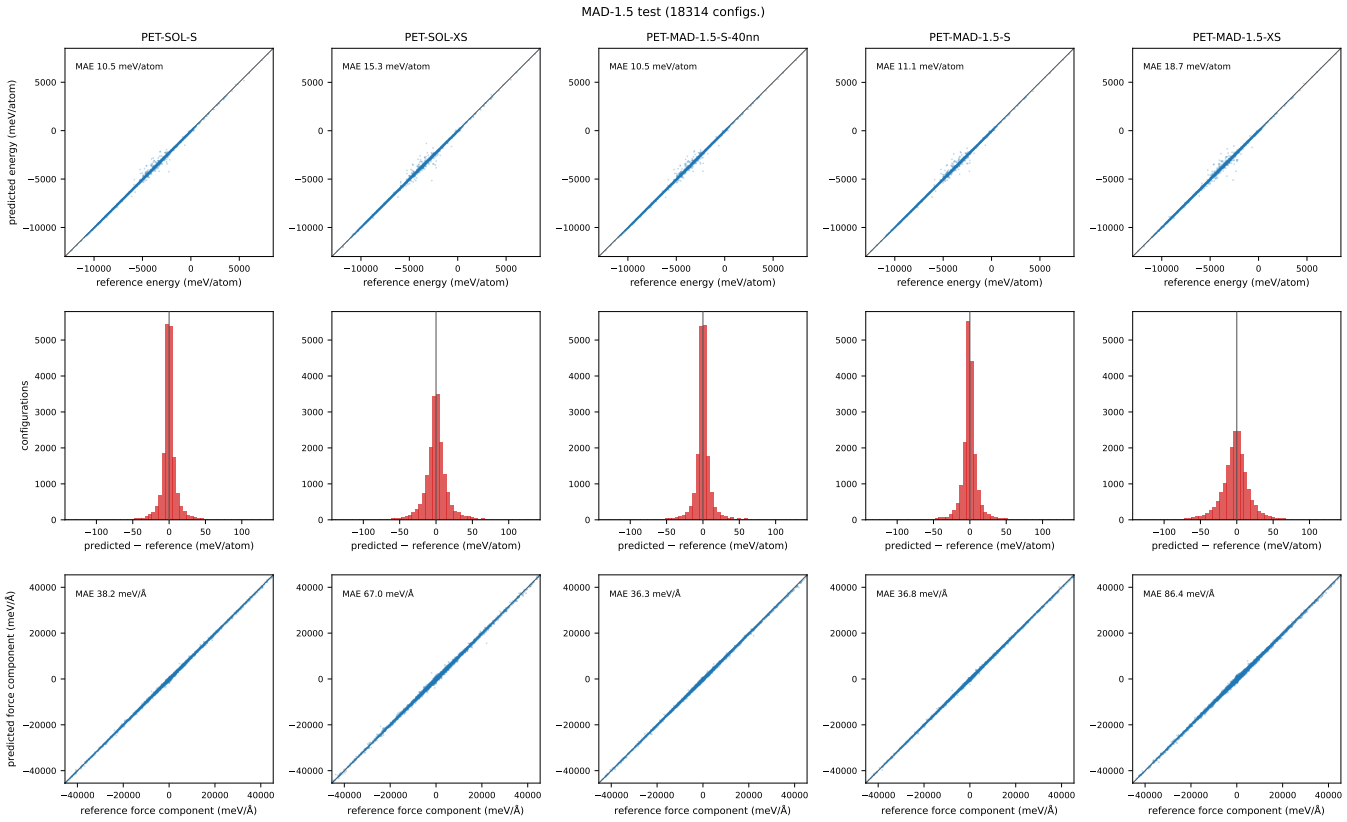


FIG. 9. As Fig. 8, for the MAD-1.5 test set (18 314 configurations).

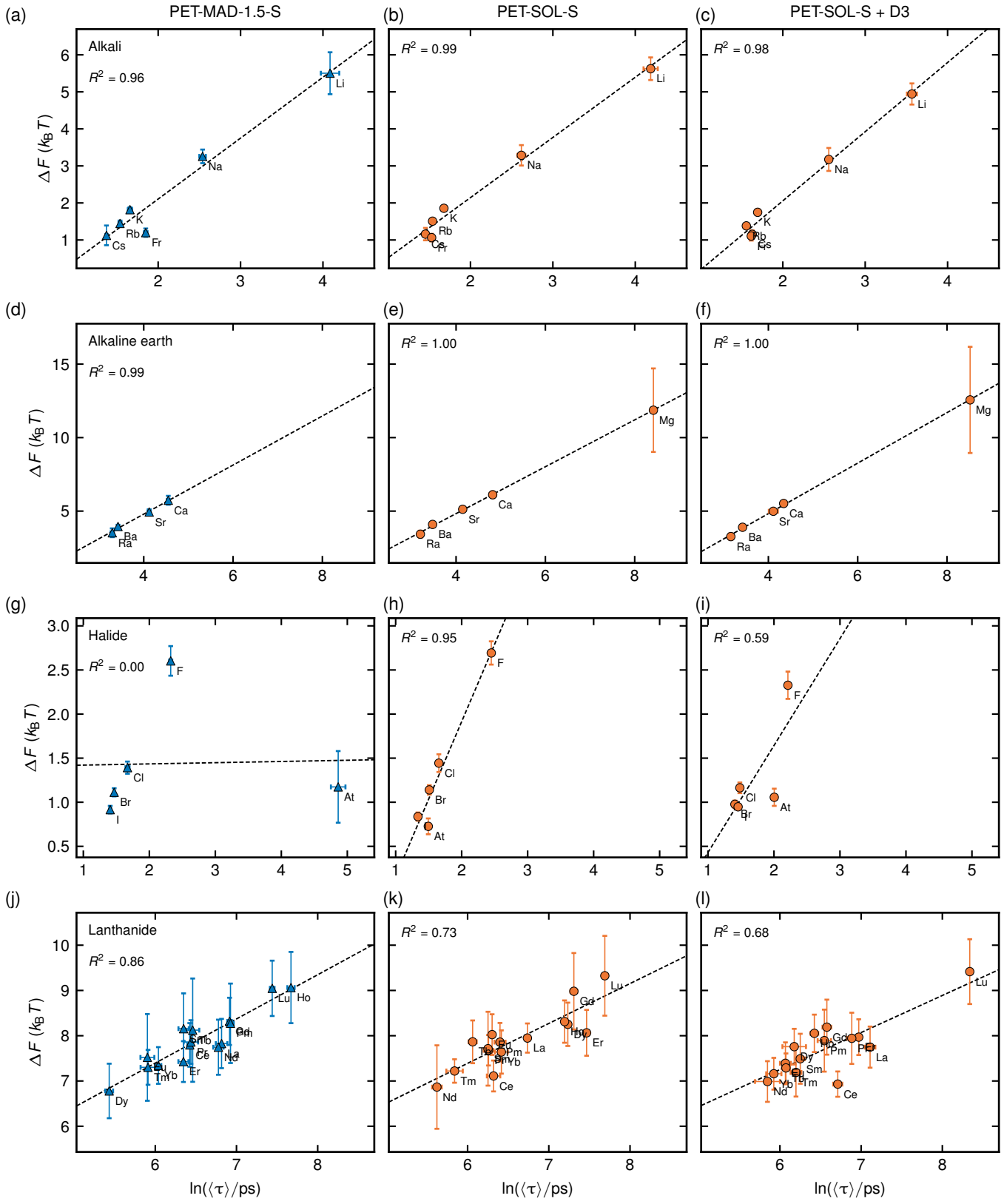


FIG. 10. First-shell free-energy barrier ΔF against $\ln(\langle\tau\rangle/\mathrm{ps})$ for the alkali metals (a–c), alkaline-earth metals (d–f), halides (g–i) and lanthanides (j–l), computed with PET-MAD-1.5-S (left column), PET-SOL-S (middle) and PET-SOL-S + D3 (right). Axes are shared within each row. Dashed lines are linear least-squares fits with the given R^2. Vertical error bars are the LLPR committee uncertainty of ΔF; horizontal error bars are the standard error of $\ln\langle\tau\rangle$ over five 1 ns blocks, which is smaller than the symbols for most ions. Be^{2+} is omitted for all models, as no water is found at its first RDF minimum within 5 ns and ΔF is not finite; the same holds for Mg^{2+} with PET-MAD-1.5-S. For PET-MAD-1.5-S, At^- is not shown in panel (g), as its first-shell water coordination differs qualitatively from that of the other halides and its $\langle\tau\rangle$ lies far outside their range. The residence time of Mg^{2+} approaches the length of a single block, so its error bar is indicative only.

E. First-shell properties across ion series

TABLE XI: First-shell ion–O properties across the three MLIP variants: position $r_{\max}$ and height $g_{\max}$ of the first RDF peak (the minimum of the PMF, $\arg\min F$), position $r_{\min}$ and height $g_{\min}$ of the following first minimum (the barrier top), $\Delta F = k_{\mathrm{B}}T\ln(g_{\max}/g_{\min})$, the coordination number integrated to $r_{\min}$, the inflection-point shell boundary r_{c} used for the residence time, and the mean residence time $\langle\tau\rangle$ with its standard error over five 1 ns blocks. $\langle\tau\rangle > 2.5$ ns marks residence times beyond half of the 5 ns trajectory, which are lower bounds. ΔF is not finite (–) where $g_{\min} = 0$.

Quantity	PET-MAD-1.5-S	PET-SOL-S	PET-SOL-S + D3
Alkali			
Li			
$r_{\max} = \arg\min F$ (Å)	1.95 ± 0.01	1.94 ± 0.01	1.94 ± 0.01
$g_{\max}$	9.15	9.16	8.21
$r_{\min}$ (Å)	2.69 ± 0.06	2.74 ± 0.05	2.74 ± 0.06
$g_{\min}$	0.0373	0.0331	0.0586
ΔF ($k_{\mathrm{B}}T$)	5.50 ± 0.57	5.62 ± 0.31	4.94 ± 0.29
CN (to $r_{\min}$)	4.1	4.1	4.1
r_{c}, inflection (Å)	2.09	2.09	2.09
$\langle\tau\rangle$ (ps)	42.8 ± 6.3	45.4 ± 4.8	26.2 ± 1.4
Na			
$r_{\max} = \arg\min F$ (Å)	2.35 ± 0.02	2.35 ± 0.01	2.36 ± 0.01
$g_{\max}$	5.60	5.67	5.30
$r_{\min}$ (Å)	3.18 ± 0.06	3.19 ± 0.05	3.17 ± 0.05
$g_{\min}$	0.216	0.212	0.222
ΔF ($k_{\mathrm{B}}T$)	3.26 ± 0.18	3.29 ± 0.28	3.17 ± 0.31
CN (to $r_{\min}$)	5.5	5.5	5.7
r_{c}, inflection (Å)	2.53	2.53	2.53
$\langle\tau\rangle$ (ps)	7.87 ± 0.22	8.69 ± 0.29	8.07 ± 0.13
K			
$r_{\max} = \arg\min F$ (Å)	2.78 ± 0.01	2.77 ± 0.02	2.76 ± 0.02
$g_{\max}$	3.49	3.58	3.41
$r_{\min}$ (Å)	3.63 ± 0.06	3.64 ± 0.08	3.55 ± 0.06
$g_{\min}$	0.567	0.56	0.594
ΔF ($k_{\mathrm{B}}T$)	1.82 ± 0.06	1.86 ± 0.06	1.75 ± 0.08
CN (to $r_{\min}$)	7.0	6.9	7.2
r_{c}, inflection (Å)	2.98	2.97	2.97
$\langle\tau\rangle$ (ps)	3.24 ± 0.06	3.39 ± 0.03	3.26 ± 0.05
Rb			
$r_{\max} = \arg\min F$ (Å)	2.96 ± 0.02	2.95 ± 0.01	2.95 ± 0.01
$g_{\max}$	2.98	3.09	2.94
$r_{\min}$ (Å)	3.93 ± 0.07	3.84 ± 0.05	3.94 ± 0.08
$g_{\min}$	0.702	0.685	0.738
ΔF ($k_{\mathrm{B}}T$)	1.45 ± 0.08	1.51 ± 0.07	1.38 ± 0.04
CN (to $r_{\min}$)	8.2	7.8	9.2
r_{c}, inflection (Å)	3.17	3.16	3.17
$\langle\tau\rangle$ (ps)	2.77 ± 0.07	2.86 ± 0.06	2.74 ± 0.08
Cs			
$r_{\max} = \arg\min F$ (Å)	3.19 ± 0.08	3.18 ± 0.03	3.19 ± 0.03
$g_{\max}$	2.52	2.58	2.51
$r_{\min}$ (Å)	4.17 ± 0.11	4.29 ± 0.09	4.37 ± 0.08
$g_{\min}$	0.82	0.81	0.793
ΔF ($k_{\mathrm{B}}T$)	1.12 ± 0.27	1.16 ± 0.17	1.15 ± 0.10
CN (to $r_{\min}$)	9.6	10.4	12.2
r_{c}, inflection (Å)	3.41	3.41	3.42

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Quantity	PET-MAD-1.5-S	PET-SOL-S	PET-SOL-S + D3
$\langle\tau\rangle$ (ps)	2.30 ± 0.03	2.40 ± 0.02	2.39 ± 0.06
Fr			
$r_{\max} = \arg\min F$ (Å)	3.49 ± 0.07	3.29 ± 0.04	3.30 ± 0.04
$g_{\max}$	2.35	2.35	2.37
$r_{\min}$ (Å)	4.63 ± 0.08	4.45 ± 0.08	4.38 ± 0.07
$g_{\min}$	0.711	0.812	0.783
ΔF ($k_{\mathrm{B}}T$)	1.20 ± 0.11	1.06 ± 0.10	1.11 ± 0.12
CN (to $r_{\min}$)	14.1	11.4	12.3
r_{c}, inflection (Å)	3.81	3.55	3.58
$\langle\tau\rangle$ (ps)	3.06 ± 0.05	2.48 ± 0.06	2.58 ± 0.03
Alkaline earth			
Be			
$r_{\max} = \arg\min F$ (Å)	1.62 ± 0.01	1.62 ± 0.02	1.62 ± 0.02
$g_{\max}$	25.09	24.89	21.67
$r_{\min}$ (Å)	2.00	2.34	2.31
$g_{\min}$	0	0	0
ΔF ($k_{\mathrm{B}}T$)	–	–	–
CN (to $r_{\min}$)	4.0	4.0	4.0
r_{c}, inflection (Å)	1.73	1.74	1.73
$\langle\tau\rangle$ (ps)	> 2.5 ns	> 2.5 ns	> 2.5 ns
Mg			
$r_{\max} = \arg\min F$ (Å)	2.07 ± 0.01	2.07 ± 0.02	2.07 ± 0.02
$g_{\max}$	14.95	14.95	14.13
$r_{\min}$ (Å)	2.93	3.04 ± 0.07	3.02 ± 0.06
$g_{\min}$	0	1.1e-04	4.9e-05
ΔF ($k_{\mathrm{B}}T$)	–	11.86 ± 2.84	12.57 ± 3.61
CN (to $r_{\min}$)	6.0	6.0	6.0
r_{c}, inflection (Å)	2.20	2.20	2.20
$\langle\tau\rangle$ (ps)	1026 ± 62	949 ± 31	1856 ± 28
Ca			
$r_{\max} = \arg\min F$ (Å)	2.40 ± 0.01	2.39 ± 0.01	2.40 ± 0.01
$g_{\max}$	10.02	10.17	9.05
$r_{\min}$ (Å)	3.28 ± 0.05	3.28 ± 0.05	3.21 ± 0.05
$g_{\min}$	0.0323	0.0225	0.0363
ΔF ($k_{\mathrm{B}}T$)	5.74 ± 0.30	6.11 ± 0.23	5.52 ± 0.18
CN (to $r_{\min}$)	7.1	7.0	7.2
r_{c}, inflection (Å)	2.55	2.54	2.55
$\langle\tau\rangle$ (ps)	55.7 ± 2.1	65.2 ± 4.6	42.5 ± 1.9
Sr			
$r_{\max} = \arg\min F$ (Å)	2.59 ± 0.02	2.58 ± 0.01	2.59 ± 0.01
$g_{\max}$	8.31	8.46	7.74
$r_{\min}$ (Å)	3.43 ± 0.05	3.44 ± 0.05	3.45 ± 0.04
$g_{\min}$	0.0595	0.0503	0.0528
ΔF ($k_{\mathrm{B}}T$)	4.94 ± 0.18	5.12 ± 0.15	4.99 ± 0.15
CN (to $r_{\min}$)	7.9	7.8	8.1
r_{c}, inflection (Å)	2.75	2.74	2.75
$\langle\tau\rangle$ (ps)	32.3 ± 0.5	35.7 ± 2.7	35.9 ± 3.0
Ba			
$r_{\max} = \arg\min F$ (Å)	2.80 ± 0.01	2.79 ± 0.01	2.80 ± 0.01
$g_{\max}$	6.64	6.79	6.21
$r_{\min}$ (Å)	3.62 ± 0.06	3.63 ± 0.04	3.59 ± 0.05
$g_{\min}$	0.127	0.113	0.126
ΔF ($k_{\mathrm{B}}T$)	3.96 ± 0.11	4.09 ± 0.10	3.90 ± 0.11
CN (to $r_{\min}$)	8.7	8.6	8.9
r_{c}, inflection (Å)	2.97	2.96	2.97

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Quantity	PET-MAD-1.5-S	PET-SOL-S	PET-SOL-S + D3
$\langle\tau\rangle$ (ps)	17.2 ± 0.5	18.7 ± 0.1	18.0 ± 0.4
Ra			
$r_{\max} = \arg\min F$ (Å)	2.89 ± 0.02	2.91 ± 0.02	2.92 ± 0.02
$g_{\max}$	6.04	5.93	5.49
$r_{\min}$ (Å)	3.78 ± 0.07	3.83 ± 0.05	3.79 ± 0.06
$g_{\min}$	0.178	0.192	0.209
ΔF ($k_{\mathrm{B}}T$)	3.53 ± 0.30	3.43 ± 0.24	3.27 ± 0.22
CN (to $r_{\min}$)	9.4	9.4	9.8
r_{c}, inflection (Å)	3.08	3.10	3.11
$\langle\tau\rangle$ (ps)	15.2 ± 0.9	13.7 ± 0.3	12.9 ± 0.3
Halide			
F			
$r_{\max} = \arg\min F$ (Å)	2.62 ± 0.01	2.61 ± 0.01	2.61 ± 0.01
$g_{\max}$	4.41	4.50	4.18
$r_{\min}$ (Å)	3.32 ± 0.05	3.29 ± 0.05	3.22 ± 0.05
$g_{\min}$	0.326	0.304	0.408
ΔF ($k_{\mathrm{B}}T$)	2.60 ± 0.17	2.69 ± 0.13	2.33 ± 0.15
CN (to $r_{\min}$)	5.1	5.1	5.3
r_{c}, inflection (Å)	2.78	2.78	2.78
$\langle\tau\rangle$ (ps)	7.15 ± 0.17	7.81 ± 0.17	6.19 ± 0.10
Cl			
$r_{\max} = \arg\min F$ (Å)	3.15 ± 0.02	3.15 ± 0.01	3.14 ± 0.02
$g_{\max}$	2.83	2.90	2.75
$r_{\min}$ (Å)	3.87 ± 0.05	3.79 ± 0.07	3.89 ± 0.08
$g_{\min}$	0.704	0.684	0.859
ΔF ($k_{\mathrm{B}}T$)	1.39 ± 0.07	1.44 ± 0.10	1.16 ± 0.06
CN (to $r_{\min}$)	6.9	6.5	8.0
r_{c}, inflection (Å)	3.34	3.35	3.33
$\langle\tau\rangle$ (ps)	3.60 ± 0.08	3.57 ± 0.04	2.96 ± 0.06
Br			
$r_{\max} = \arg\min F$ (Å)	3.33 ± 0.02	3.32 ± 0.02	3.31 ± 0.02
$g_{\max}$	2.54	2.53	2.49
$r_{\min}$ (Å)	4.07 ± 0.07	4.09 ± 0.05	4.56 ± 0.10
$g_{\min}$	0.833	0.81	0.936
ΔF ($k_{\mathrm{B}}T$)	1.11 ± 0.05	1.14 ± 0.05	0.98 ± 0.04
CN (to $r_{\min}$)	7.8	7.8	13.1
r_{c}, inflection (Å)	3.52	3.52	3.51
$\langle\tau\rangle$ (ps)	2.89 ± 0.03	3.02 ± 0.03	2.57 ± 0.03
I			
$r_{\max} = \arg\min F$ (Å)	3.57 ± 0.02	3.58 ± 0.03	3.59 ± 0.03
$g_{\max}$	2.33	2.18	2.23
$r_{\min}$ (Å)	4.33 ± 0.07	4.34 ± 0.06	5.23 ± 0.09
$g_{\min}$	0.933	0.943	0.863
ΔF ($k_{\mathrm{B}}T$)	0.92 ± 0.04	0.84 ± 0.05	0.95 ± 0.03
CN (to $r_{\min}$)	9.3	9.0	19.9
r_{c}, inflection (Å)	3.76	3.79	3.79
$\langle\tau\rangle$ (ps)	2.65 ± 0.03	2.65 ± 0.04	2.35 ± 0.02
At			
$r_{\max} = \arg\min F$ (Å)	2.17 ± 0.01	3.72 ± 0.06	3.75 ± 0.05
$g_{\max}$	3.21	1.86	2.13
$r_{\min}$ (Å)	2.35 ± 0.08	5.60 ± 0.09	5.30 ± 0.08
$g_{\min}$	0.992	0.9	0.741
ΔF ($k_{\mathrm{B}}T$)	1.17 ± 0.41	0.73 ± 0.09	1.06 ± 0.10
CN (to $r_{\min}$)	1.1	21.4	20.6
r_{c}, inflection (Å)	2.26	4.02	4.23

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Quantity	PET-MAD-1.5-S	PET-SOL-S	PET-SOL-S + D3
$\langle\tau\rangle$ (ps)	95.8 ± 13.4	2.33 ± 0.01	4.18 ± 0.05
Lanthanide			
La			
$r_{\max} = \arg\min F$ (Å)	$2.56 \pm <0.01$	$2.56 \pm <0.01$	$2.56 \pm <0.01$
$g_{\max}$	11.75	12.01	10.96
$r_{\min}$ (Å)	3.48 ± 0.08	3.53 ± 0.06	3.45 ± 0.06
$g_{\min}$	0.00468	0.00424	0.00473
ΔF ($k_{\mathrm{B}}T$)	7.83 ± 0.54	7.95 ± 0.32	7.75 ± 0.45
CN (to $r_{\min}$)	9.0	8.9	9.0
r_{c}, inflection (Å)	2.71	2.71	2.70
$\langle\tau\rangle$ (ps)	248 ± 18	235 ± 26	270 ± 39
Ce			
$r_{\max} = \arg\min F$ (Å)	2.50 ± 0.01	2.56 ± 0.01	2.56 ± 0.01
$g_{\max}$	11.93	10.94	9.87
$r_{\min}$ (Å)	3.42 ± 0.06	3.48 ± 0.05	3.49 ± 0.05
$g_{\min}$	0.0049	0.00893	0.00965
ΔF ($k_{\mathrm{B}}T$)	7.80 ± 0.48	7.11 ± 0.34	6.93 ± 0.28
CN (to $r_{\min}$)	8.7	8.9	9.0
r_{c}, inflection (Å)	2.64	2.71	2.71
$\langle\tau\rangle$ (ps)	189 ± 21	128 ± 7	163 ± 26
Pr			
$r_{\max} = \arg\min F$ (Å)	2.49 ± 0.01	2.50 ± 0.01	2.51 ± 0.01
$g_{\max}$	12.10	12.29	11.17
$r_{\min}$ (Å)	3.38 ± 0.07	3.48 ± 0.05	3.40 ± 0.06
$g_{\min}$	0.00466	0.00568	0.00397
ΔF ($k_{\mathrm{B}}T$)	7.86 ± 0.48	7.68 ± 0.34	7.94 ± 0.57
CN (to $r_{\min}$)	8.7	8.7	8.9
r_{c}, inflection (Å)	2.63	2.65	2.65
$\langle\tau\rangle$ (ps)	146 ± 19	175 ± 18	201 ± 23
Nd			
$r_{\max} = \arg\min F$ (Å)	2.48 ± 0.01	2.48 ± 0.02	2.49 ± 0.02
$g_{\max}$	11.72	11.98	10.83
$r_{\min}$ (Å)	3.42 ± 0.05	3.33 ± 0.08	3.44 ± 0.06
$g_{\min}$	0.00507	0.0125	0.00999
ΔF ($k_{\mathrm{B}}T$)	7.75 ± 0.61	6.87 ± 0.92	6.99 ± 0.45
CN (to $r_{\min}$)	8.9	8.4	8.7
r_{c}, inflection (Å)	2.63	2.63	2.63
$\langle\tau\rangle$ (ps)	137 ± 10	102 ± 7	109 ± 9
Pm			
$r_{\max} = \arg\min F$ (Å)	2.45 ± 0.02	2.47 ± 0.03	2.48 ± 0.02
$g_{\max}$	12.47	12.36	11.35
$r_{\min}$ (Å)	3.37 ± 0.07	3.42 ± 0.09	3.40 ± 0.08
$g_{\min}$	0.00318	0.00478	0.00424
ΔF ($k_{\mathrm{B}}T$)	8.27 ± 0.88	7.86 ± 0.42	7.89 ± 0.68
CN (to $r_{\min}$)	8.9	8.6	8.9
r_{c}, inflection (Å)	2.60	2.62	2.62
$\langle\tau\rangle$ (ps)	173 ± 13	195 ± 40	251 ± 30
Sm			
$r_{\max} = \arg\min F$ (Å)	2.43 ± 0.02	2.44 ± 0.04	2.45 ± 0.04
$g_{\max}$	13.14	12.90	11.51
$r_{\min}$ (Å)	3.38 ± 0.09	3.33 ± 0.09	3.40 ± 0.08
$g_{\min}$	0.00377	0.00576	0.00642
ΔF ($k_{\mathrm{B}}T$)	8.16 ± 0.78	7.71 ± 0.82	7.49 ± 0.55
CN (to $r_{\min}$)	8.6	8.3	8.6
r_{c}, inflection (Å)	2.57	2.58	2.59

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Quantity	PET-MAD-1.5-S	PET-SOL-S	PET-SOL-S + D3
$\langle\tau\rangle$ (ps)	186 ± 22	174 ± 23	130 ± 11
Eu			
$r_{\max} = \arg\min F$ (Å)	2.41 ± 0.01	2.42 ± 0.01	2.43 ± 0.02
$g_{\max}$	13.04	13.47	12.03
$r_{\min}$ (Å)	3.28 ± 0.09	3.38 ± 0.07	3.26 ± 0.06
$g_{\min}$	0.00706	0.00441	0.00742
ΔF ($k_{\mathrm{B}}T$)	7.52 ± 0.96	8.02 ± 0.45	7.39 ± 0.46
CN (to $r_{\min}$)	8.2	8.2	8.6
r_{c}, inflection (Å)	2.55	2.56	2.57
$\langle\tau\rangle$ (ps)	167 ± 10	213 ± 8	161 ± 20
Gd			
$r_{\max} = \arg\min F$ (Å)	2.41 ± 0.02	2.41 ± 0.01	2.41 ± 0.02
$g_{\max}$	12.69	13.70	12.58
$r_{\min}$ (Å)	3.32 ± 0.07	3.18 ± 0.05	3.21 ± 0.07
$g_{\min}$	0.0031	0.00172	0.00349
ΔF ($k_{\mathrm{B}}T$)	8.32 ± 0.52	8.98 ± 0.84	8.19 ± 0.61
CN (to $r_{\min}$)	8.2	8.0	8.1
r_{c}, inflection (Å)	2.55	2.54	2.54
$\langle\tau\rangle$ (ps)	200 ± 34	424 ± 46	297 ± 37
Tb			
$r_{\max} = \arg\min F$ (Å)	2.39 ± 0.02	2.39 ± 0.01	2.40 ± 0.02
$g_{\max}$	12.89	13.88	12.10
$r_{\min}$ (Å)	3.27 ± 0.07	3.33 ± 0.07	3.25 ± 0.08
$g_{\min}$	0.00382	0.00533	0.00827
ΔF ($k_{\mathrm{B}}T$)	8.12 ± 1.14	7.87 ± 0.47	7.29 ± 0.32
CN (to $r_{\min}$)	8.2	8.2	8.6
r_{c}, inflection (Å)	2.54	2.53	2.55
$\langle\tau\rangle$ (ps)	153 ± 16	151 ± 6	118 ± 12
Dy			
$r_{\max} = \arg\min F$ (Å)	2.40 ± 0.01	2.41 ± 0.01	2.41 ± 0.01
$g_{\max}$	12.49	14.15	12.78
$r_{\min}$ (Å)	3.33 ± 0.06	3.28 ± 0.08	3.21 ± 0.07
$g_{\min}$	0.0142	0.00369	0.00546
ΔF ($k_{\mathrm{B}}T$)	6.78 ± 0.60	8.25 ± 0.48	7.76 ± 0.39
CN (to $r_{\min}$)	7.8	8.0	8.2
r_{c}, inflection (Å)	2.55	2.55	2.55
$\langle\tau\rangle$ (ps)	114 ± 5	300 ± 41	218 ± 23
Ho			
$r_{\max} = \arg\min F$ (Å)	2.39 ± 0.01	2.38 ± 0.01	2.38 ± 0.01
$g_{\max}$	13.21	13.30	11.80
$r_{\min}$ (Å)	3.28 ± 0.07	3.38 ± 0.07	3.21 ± 0.05
$g_{\min}$	0.00153	0.00326	0.00376
ΔF ($k_{\mathrm{B}}T$)	9.06 ± 0.79	8.31 ± 0.47	8.05 ± 0.41
CN (to $r_{\min}$)	8.0	8.0	8.0
r_{c}, inflection (Å)	2.53	2.52	2.52
$\langle\tau\rangle$ (ps)	332 ± 39	247 ± 40	204 ± 26
Er			
$r_{\max} = \arg\min F$ (Å)	2.37 ± 0.01	2.37 ± 0.01	2.37 ± 0.01
$g_{\max}$	12.12	13.15	12.25
$r_{\min}$ (Å)	3.28 ± 0.08	3.28 ± 0.09	3.26 ± 0.06
$g_{\min}$	0.0072	0.00414	0.00424
ΔF ($k_{\mathrm{B}}T$)	7.43 ± 0.45	8.06 ± 0.51	7.97 ± 0.39
CN (to $r_{\min}$)	8.0	8.0	8.0
r_{c}, inflection (Å)	2.52	2.52	2.51
$\langle\tau\rangle$ (ps)	173 ± 13	274 ± 25	214 ± 13

continued on next page

(continued)

Quantity	PET-MAD-1.5-S	PET-SOL-S	PET-SOL-S + D3
Tm			
$r_{\max} = \arg\min F$ (Å)	2.36 ± 0.01	2.36 ± 0.01	2.37 ± 0.01
$g_{\max}$	12.21	12.48	11.28
$r_{\min}$ (Å)	3.28 ± 0.08	3.34 ± 0.07	3.21 ± 0.05
$g_{\min}$	0.00823	0.00913	0.00851
ΔF ($k_{\mathrm{B}}T$)	7.30 ± 0.39	7.22 ± 0.26	7.19 ± 0.53
CN (to $r_{\min}$)	7.8	7.6	7.8
r_{c}, inflection (Å)	2.51	2.51	2.51
$\langle\tau\rangle$ (ps)	128 ± 10	112 ± 8	135 ± 10
Yb			
$r_{\max} = \arg\min F$ (Å)	2.44 ± 0.01	2.45 ± 0.01	2.45 ± 0.01
$g_{\max}$	12.61	12.72	11.57
$r_{\min}$ (Å)	3.33 ± 0.06	3.33 ± 0.05	3.26 ± 0.07
$g_{\min}$	0.00817	0.00611	0.00899
ΔF ($k_{\mathrm{B}}T$)	7.34 ± 0.40	7.64 ± 0.48	7.16 ± 0.35
CN (to $r_{\min}$)	8.0	8.0	8.1
r_{c}, inflection (Å)	2.58	2.59	2.59
$\langle\tau\rangle$ (ps)	207 ± 16	246 ± 31	190 ± 37
Lu			
$r_{\max} = \arg\min F$ (Å)	2.30 ± 0.01	$2.31 \pm <0.01$	$2.31 \pm <0.01$
$g_{\max}$	15.43	15.69	14.48
$r_{\min}$ (Å)	3.18 ± 0.09	3.38 ± 0.06	3.21 ± 0.10
$g_{\min}$	0.00182	0.0014	0.00118
ΔF ($k_{\mathrm{B}}T$)	9.05 ± 0.61	9.32 ± 0.88	9.42 ± 0.72
CN (to $r_{\min}$)	7.9	7.9	8.0
r_{c}, inflection (Å)	2.44	2.44	2.44
$\langle\tau\rangle$ (ps)	343 ± 12	439 ± 19	627 ± 46

F. Composition of the finite-concentration electrolyte boxes

TABLE XII. Ion and water counts of the ≈ 1 M finite-concentration boxes (Sec. II D 3).

Salt	Model	n_{cation}	n_{anion}	n_{water}
$CaCl_2$	PET-MAD-1.5-S	10	20	506
	PET-SOL-S	10	20	506
	PET-SOL-S + D3	9	18	509
NaCl	PET-MAD-1.5-S	10	10	539
	PET-SOL-S	10	10	509
	PET-SOL-S + D3	9	9	511
CsCl	PET-MAD-1.5-S	10	10	496
	PET-SOL-S	10	10	496
	PET-SOL-S + D3	9	9	500

G. Water–water hydrogen-bond dynamics

The persistence of the water–water hydrogen-bond network was quantified with the intermittent hydrogen-bond correlation function of Luzar and Chandler [99]. Two water molecules are taken as hydrogen bonded if their oxygen–oxygen distance is at most 3.5 Å and the angle between the donor O–H bond and the O$\cdots$O vector is at most 30° [99]; hydrogens are assigned to their donor oxygen within 1.2 Å. Candidate bonds were detected with `MDAnalysis`'s `HydrogenBondAnalysis` [53, 100] and the angular criterion was applied to the minimum-image positions of every analysed frame. For each donor–acceptor pair a Boolean indicator $h(t)$ was sampled every 0.1 ps along the production trajectories. Each bond present at a time origin t_0 is assigned to a population: the cation (anion) shell if the oxygen of either water lies within the first minimum of the cation–O (anion–O) radial distribution function of the same system, and bulk if neither does. The assignment is made only at t_0, so that later exchange of a water out of the shell does not terminate $h(t)$. The correlation function of each population,

$$C_{\mathrm{HB}}(t) = \frac{\sum h(t_0)\, m(t_0)\, h(t_0 + t)}{\sum h(t_0)\, m(t_0)}, \tag{21}$$

with $m(t_0) = 1$ for bonds in that population, is accumulated over all pairs, time origins and the four replicas before dividing. We report the relaxation time $\tau_{\mathrm{HB}} = \int_0^{100\,\mathrm{ps}} C_{\mathrm{HB}}(t)\,\mathrm{d}t$, which does not depend on the choice of a fitting function, with the standard error of the per-replica integrals, and quantify ion effects by the ratio $\tau_{\mathrm{HB}}^{\mathrm{shell}}/\tau_{\mathrm{HB}}^{\mathrm{bulk}}$.

TABLE XIII. Hydration-shell dynamics of the ≈ 1 M solutions: hydrogen-bond relaxation time in the first hydration shell of the cation and of Cl^- relative to bulk water, $\tau_{\mathrm{HB}}^{\mathrm{shell}}/\tau_{\mathrm{HB}}^{\mathrm{bulk}}$ (Sec. V G), and the mean residence time $\langle\tau\rangle$ of water in the first shell of the cation and of Cl^-, computed as in the dilute limit (Sec. II E 3) with the shell cutoff taken from the RDFs of each solution. Uncertainties are standard errors of the per-replica ratios (τ_{HB}) and over the ion instances ($\langle\tau\rangle$). As for the reorientation (Sec. II E 5), the PET-MAD-1.5-S NaCl replica containing an autoionised H_3O^+/OH^- pair is excluded from $\langle\tau\rangle$.

		$\tau_{\mathrm{HB}}^{\mathrm{shell}}/\tau_{\mathrm{HB}}^{\mathrm{bulk}}$		$\langle\tau\rangle$ (ps)	
Salt	Model	Cation shell	Cl^- shell	Cation shell	Cl^- shell
$CaCl_2$ (Ca^{2+})	PET-MAD-1.5-S	1.15 ± 0.01	1.00 ± <0.01	95.8 ± 1.4	5.6 ± <0.1
	PET-SOL-S	1.17 ± <0.01	1.00 ± <0.01	99.8 ± 1.5	5.7 ± <0.1
	PET-SOL-S + D3	1.26 ± 0.01	1.02 ± <0.01	79.2 ± 1.2	5.2 ± <0.1
CsCl (Cs^+)	PET-MAD-1.5-S	0.97 ± <0.01	0.96 ± <0.01	4.2 ± <0.1	5.0 ± <0.1
	PET-SOL-S	0.96 ± <0.01	0.95 ± <0.01	4.3 ± <0.1	5.0 ± <0.1
	PET-SOL-S + D3	1.00 ± <0.01	0.97 ± <0.01	5.0 ± <0.1	4.5 ± <0.1
NaCl (Na^+)	PET-MAD-1.5-S	0.94 ± <0.01	0.97 ± 0.01	13.5 ± 0.1	5.1 ± <0.1
	PET-SOL-S	0.94 ± <0.01	0.95 ± <0.01	13.3 ± 0.1	5.2 ± <0.1
	PET-SOL-S + D3	1.02 ± <0.01	0.98 ± <0.01	13.9 ± 0.1	4.6 ± <0.1

Water–water hydrogen-bond dynamics (SI Sec. V G) are slowed only in the first shell of Ca^{2+}, by 15–26 % relative to bulk water, whereas the shells of Na^+, Cs^+ and Cl^- stay within 7 % of bulk in all three models (Table XIII).

H. Multivalent ions

A machine-learning potential trained without an explicit notion of formal oxidation state could, in principle, learn a spuriously stoichiometry-dependent description of a multivalent cation: for example, representing Fe differently depending on how many chloride counter-ions happen to be nearby, rather than on a single, continuous potential-energy surface. In order to investigate this, we simulated a small Fe–Cl series — one Fe cation together with $n = 0$ to 4 Cl^- counter-ions packed with 256 water molecules into the same box (Sec. II D 1) — and computed the Fe–O RDF for each n with the protocol of Sec. II E 1.

Figure 11 shows the result. For every n, Fe is octahedrally coordinated by six water molecules ($N(r) = 6.0$ beyond the first peak) and no Cl^- enters its first shell. The Fe–O peak lies between $r_{max} = 2.01$ Å ($n = 0$) and 1.94 Å ($n = 4$), close to the experimental Fe–O distance of hydrated Fe^{3+} (≈ 2.0 Å) and clearly below that of Fe^{2+} (≈ 2.1 Å) [83]. Fe is therefore described as Fe^{3+} for every n, including $n = 0$, where the box contains no counter-ion at all: the model, which carries no charge information, does not infer the oxidation state from the counter-ions but retains the Fe^{3+} description learned from the training data, in which Fe appears with formal charge +3. Adding Cl^- shortens the Fe–O distance only gradually, by ≈ 0.017 Å per counter-ion and without any discontinuity, as expected for a single, consistent potential-energy surface with a small electrostatic effect of the surrounding anions. Conversely, the model cannot describe Fe^{2+} without additional training data.

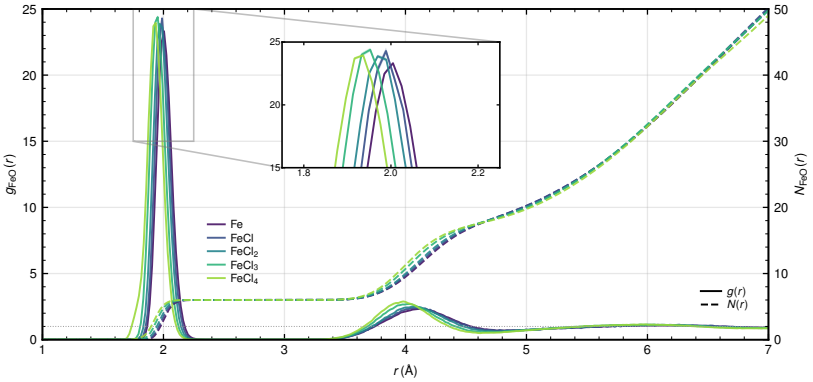


FIG. 11. Fe–O radial distribution function $g(r)$ (solid) and cumulative coordination number $N(r)$ (dashed) for a single Fe cation with $n = 0, 1, 2, 3, 4$ Cl^- counter-ions in water, computed with PET-SOL-S.

I. Uncertainty quantification

Throughout this work, model uncertainties were computed from the Last Layer Prediction Rigidity formalism [55]. This approach treats the last layer as Bayesian Linear Regression, sampling weights from a normal distribution around the optimized weights with covariance $\alpha^2 \left(\mathbf{F}^\top \mathbf{F} + \varsigma^2 \mathbf{I}\right)^{-1}$, where $\mathbf{F}$ is the matrix of last-layer features of the training structures, ς is a regularization parameter, and α is a scalar calibration factor fitted on a held-out validation set by minimizing the negative log-likelihood. Drawing M weight sets yields, for every frame, an ensemble of energy predictions $\{V_m\}_{m=1}^{M}$, whose spread reflects the uncertainty of the model at that configuration. To propagate this uncertainty to ensemble averages, the trajectory generated with the reference potential $\bar{V}$ was reweighted to each ensemble member following Ref. [56].

$$p_i = \frac{\langle \delta w_i \rangle_{\bar{V}}}{\langle w_i \rangle_{\bar{V}}} \tag{22}$$

, where the weights for each committee member i of configuration q are defined as $w_i(q) = e^{-\beta(V_i(q) - \bar{V(q)})}$, $\bar{V}$ is the expected potential energy and δ is short for the Dirac delta $\delta(s(q) - s)$. The authors propose a first order cumulant expansion in order to facilitate sampling of the exponentiated potential difference. When dealing with distributions (such as the RDF) that approach is not applicable, given the Gaussian error assumption. We thus perform the cumulant expansion in log space and operate on the Potential of Mean Force (PMF). Changing the measure to space restricted by s we find

$$p_i(s) = \frac{\langle w_i \rangle_{\bar{V}|s} \langle \delta \rangle_{\bar{V}}}{\langle w_i \rangle_{\bar{V}}}. \tag{23}$$

In log space this becomes

$$\log p_i(s) = \log p(s) + \log \langle w_i \rangle_{\bar{V}|s} - \log \langle w_i \rangle_{\bar{V}} \tag{24}$$

, where we identify $\beta F_i(s) = -\log p_i(s)$ which we can expand to first order

$$\log p_i(s) \approx \log p(s) - \beta \langle \Delta V_i \rangle_{\bar{V}|s} + \beta \langle \Delta V_i \rangle_{\bar{V}}. \tag{25}$$

Dropping terms that don't depend on s, the PMF of committee member i can be estimated as

$$F_i(s) \approx F(s) + \langle \Delta V_i \rangle_{\bar{V}|s}\,. \tag{26}$$

supplies an ensemble of energy predictions allowing to estimate energy uncertainty of the total structure.

J. Bulk water structure

All simulations in this work have been performed at an elevated temperature of 330, to emulate structure at ambient conditions. SCAN potentials are known to over structure water at RT, and report a 30K increase in melting temperature. To test this choice, we compare bulk water RDFs at both 300 K and 330 K. In order to match experimental $g_{\mathrm{HH}}(r)$ and $g_{\mathrm{OH}}(r)$ nuclear quantum effects are included through path-integral molecular dynamics simulations (Figure 12). Trajectories were obtained from TODO xxx production length NpT simulations, after equilibration? The PIGLET [101, 102] thermostat was adopted to accelerate the simulation by decreasing the need for the number of ring-polymer beads to 8. Without inclusion of dispersion, our models clearly over structure water at 300 K. Simulating at 330 K gives the expected O–O RDF. Similarly to ion–O RDFs D3 dispersion leads to less depletion after the first solvation shell and a weaker second solvation shell. This aligns well, with the unphysically high density of bulk water of PET-SOL-S+D3 4.

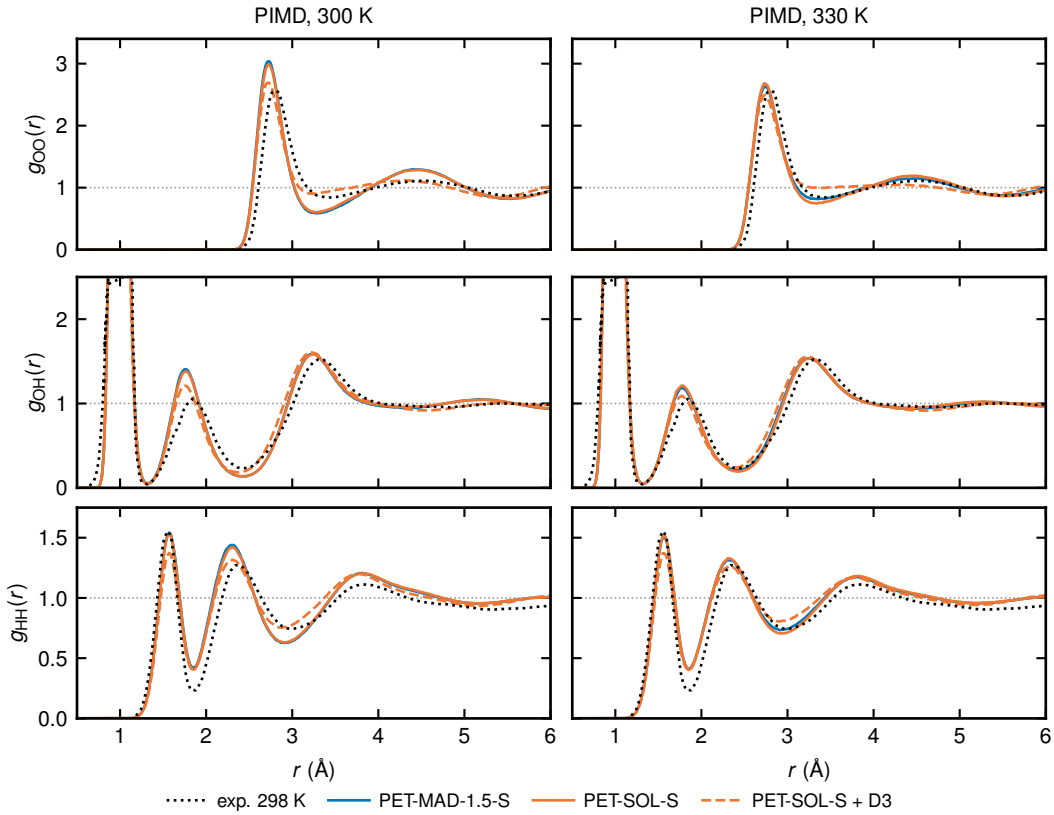


FIG. 12. Quantum bulk-water radial distribution functions against experiment, produced at (a) 300 K and (b) 330 K, for PET-MAD-1.5-S and PET-SOL-S, the latter with and without the D3 dispersion correction. Top: $g_{OO}(r)$. Middle: $g_{OH}(r)$. Bottom: $g_{HH}(r)$.